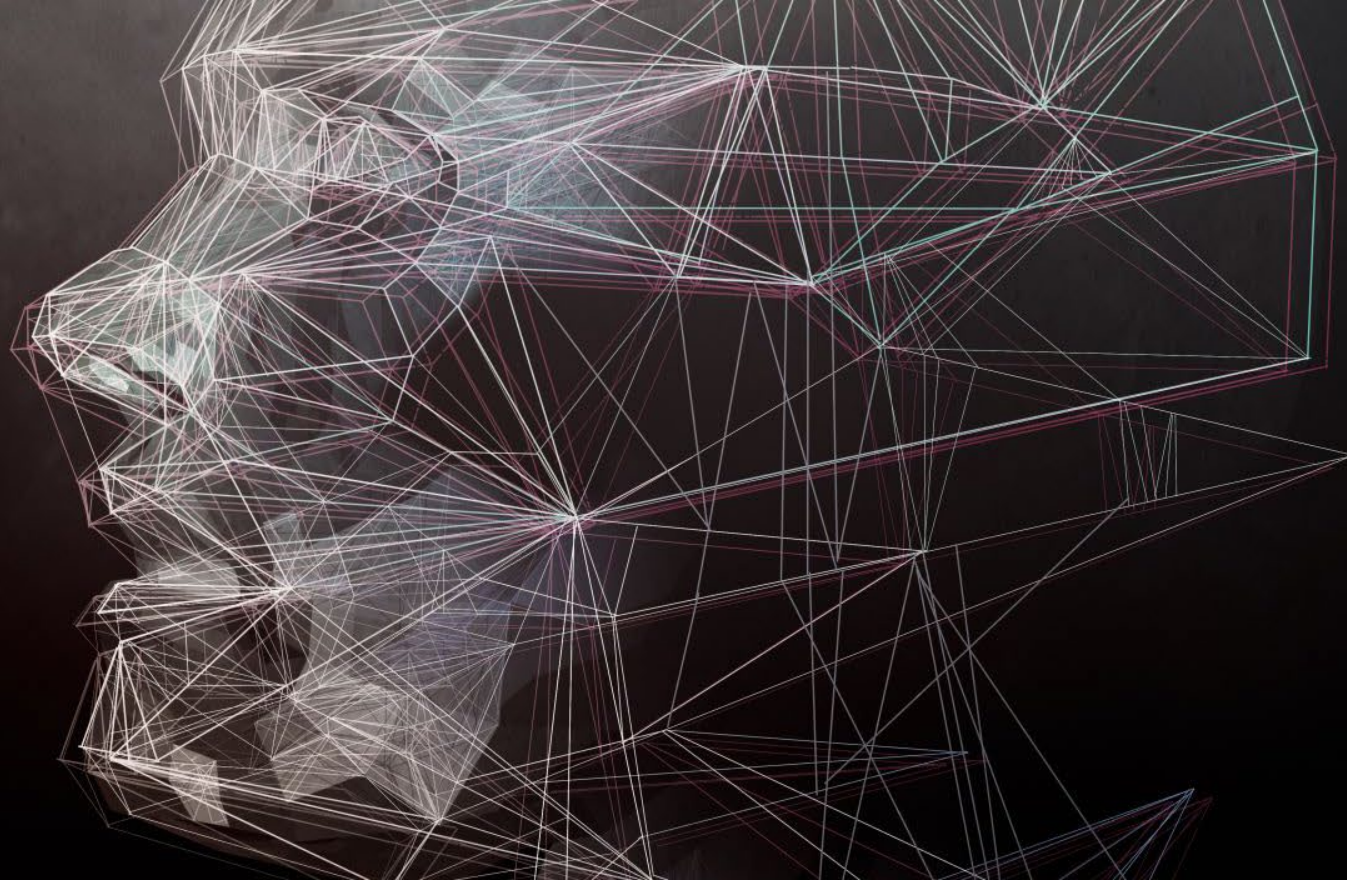
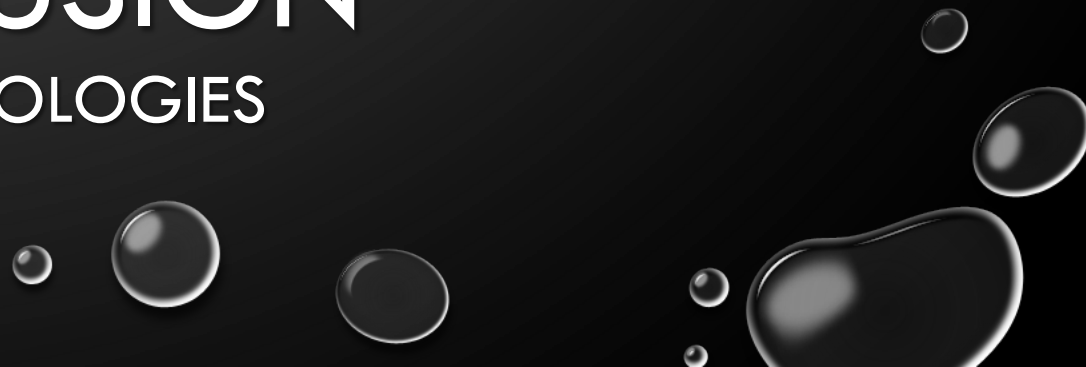




STABLE DIFFUSION

UNDERLYING TECHNOLOGIES



Slides are broadly classified under two topics :

“ The Understanding Years ” & “ The Years of Generation ”

◆ AI Overview

The "stage of understanding" and "stage of generating" refer to distinct phases in various cognitive and creative processes. Understanding involves acquiring knowledge and grasping concepts, while generating involves producing new ideas, solutions, or content. These stages often interact and can be part of larger frameworks like the creative process, design thinking, or learning models. [🔗](#)



The background of the slide is a dense, overlapping field of 3D-rendered numbers (0-9) in two colors: a light grey and a mustard yellow. The numbers are scattered across the frame, creating a sense of depth and complexity. In the top-left and bottom-right corners, there are several translucent, realistic-looking bubbles of varying sizes, some with highlights and shadows, adding a soft, organic feel to the otherwise geometric and digital background.

THE UNDERSTANDING YEARS

(centred around Neural Networks development for Vision Research & Generative AI)

SELF-ORGANIZING MAPS (SOM)

An SOM is a type of artificial neural network but is trained using competitive learning rather than the error-correction learning (e.g., backpropagation with gradient descent) used by other artificial neural networks. The SOM was introduced by the Finnish professor Teuvo Kohonen in the 1980s and therefore is sometimes called a Kohonen map or Kohonen network.

Bibliography of Self-Organizing Map (SOM) Papers: 1981–1997

Samuel Kaski†, Jari Kangas‡, Teuvo Kohonen†

† Helsinki University of Technology, Neural Networks Research Centre,
P.O. Box 2200, FIN-02015 HUT, FINLAND

‡ Nokia Research Center, P.O. Box 100, FIN-33721 Tampere, FINLAND

Abstract

The Self-Organizing Map (SOM) algorithm has attracted an ever increasing amount of interest among researchers and practitioners in a wide variety of fields. The SOM and a variant of it, the LVQ, have been analyzed extensively, a number of variants of them have been developed and, perhaps most notably, they have been applied extensively within fields ranging from engineering sciences to medicine, biology, and economics. We have collected a comprehensive list of 3343 scientific papers that use the algorithms, have benefited from them, or contain analyses of them. The list is intended to serve as a source for literature surveys. We have provided both a thematic and a keyword index to help finding articles of interest.

1 INTRODUCTION

The Self-Organizing Map algorithm [1530, 1537, 1593] was introduced in 1981. The earliest applications were mainly in engineering tasks. Later the algorithm has become progressively more accepted as a standard data analysis method in a wide variety of fields that can utilize unsupervised learning: clustering, visualization, data organization, characterization, and exploration. The variant called Learning Vector Quantization (LVQ) has additionally been used extensively in supervised tasks, especially classification and supervised pattern recognition.

Many of the papers on SOM analyze the method or present variants or generalizations of it. Most of the papers, however, apply the method or its variants in fields ranging from engineering (including image and signal processing and recognition, telecommunications, process monitoring and control, and robotics) and natural sciences to medicine, humanities, economics and mathematics. The definitive reference to the state of the art in SOMs is [1593].

1.1 Collection Method

We have been collecting a bibliography of scientific papers on SOM already for many years. Our criterion in selecting papers has been that they should either use or analyze the SOM, or benefit from it in some other manner. Our intention has been to exclude papers that merely refer to the algorithm.

Several methods have been used in collecting the bibliography. We have added references to papers that have appeared in the journals and conference proceedings that we personally follow. In addition, several authors have kindly helped us by sending us bibliographies on their own papers. Finally, we have made

CNN

Convolutional Neural Network

Developed by Yann LeCun and his team, LeNet-5 was specifically designed for handwritten digit recognition. It built upon the Neocognitron by incorporating supervised learning with backpropagation and using techniques like max-pooling, which became a standard component in CNN architectures.

Although CNNs were invented in the 1980s, their breakthrough is in the 2000s where papers started reporting fast implementations on graphics processing units (GPUs)

Communicated by Dana Ballard

Backp
Recog

Y. LeCun
B. Boser
J. S. Denker
D. Henderson
R. E. Hinton
W. Hukins
L. D. Jackel
AT&T Bell Laboratories

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1 Intro

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Published as a conference paper at ICLR 2015

VERY DEEP CONVOLUTIONAL NETWORKS FOR LARGE-SCALE IMAGE RECOGNITION

Karen Simonyan* & Andrew Zisserman*

Visual Geometry Group, Department of Engineering Science, University of Oxford
{karen, az}@robots.ox.ac.uk

ABSTRACT

In this work we investigate the effect of the convolutional network depth on its accuracy in the large-scale image recognition setting. Our main contribution is a thorough evaluation of networks of increasing depth using an architecture with very small (3×3) convolution filters, which shows that a significant improvement on the prior-art configurations can be achieved by pushing the depth to 16–19 weight layers. These findings were the basis of our ImageNet Challenge 2014 submission, where our team secured the first and the second places in the localisation and classification tracks respectively. We also show that our representations generalise well to other datasets, where they achieve state-of-the-art results. We have made our two best-performing ConvNet models publicly available to facilitate further research on the use of deep visual representations in computer vision.

1 INTRODUCTION

Convolutional networks (ConvNets) have recently enjoyed a great success in large-scale image and video recognition (Krizhevsky et al., 2012; Zeiler & Fergus, 2013; Sermanet et al., 2014; Simonyan & Zisserman, 2014) which has become possible due to the large public image repositories such as ImageNet (Deng et al., 2009), and high-performance computing systems, such as GPUs and large-scale distributed clusters (Dean et al., 2012). In particular, an important role in the advance of deep visual recognition architectures has been played by the ImageNet Large-Scale Visual Recognition Challenge (ILSVRC) (Russakovsky et al., 2014), which has served as a testbed for a few years for large-scale image classification systems, from high-dimensional shallow feature engines (Perrotin et al., 2010) (the winner of ILSVRC-2011) to deep ConvNets (Krizhevsky et al., 2012) (the winner of ILSVRC-2012).

As ConvNets becoming more of a commodity in the computer vision field, a number of attempts have been made to improve the original architecture of Krizhevsky et al. (2012) in a bid to achieve better accuracy. For instance, the best-performing submissions to the ILSVRC-2013 (Zeiler & Fergus, 2013; Sermanet et al., 2014) utilised smaller receptive window size and smaller stride of the first convolutional layer. Another line of improvements dealt with training and testing the networks densely over the whole image and over multiple scales (Sermanet et al., 2014; Howard, 2014). In this paper, we address another important aspect of ConvNet architecture design – its depth. To this end, we fix other parameters of the architecture, and steadily increase the depth of the network by adding more convolutional layers, which is feasible due to the use of very small (3×3) convolution filters in all layers.

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AlexNet is widely credited with igniting the deep learning revolution. Its success demonstrated the effectiveness of deep neural networks trained on GPUs, leading to rapid progress across multiple domains of artificial intelligence beyond computer vision. The techniques and momentum generated by AlexNet helped shape the development of modern natural language processing models, including large-scale transformer-based models such as BERT and GPT, which power tools like ChatGPT.

ALEXNET

AlexNet, a similar GPU-based CNN by Alex et al. won the ImageNet Large Scale Visual Challenge 2012. It was an early catalytic AI boom.

Alex Krizhevsky, Ilya Sutskever, Geoffrey Hilton sold their company DNNResearch in 2013 and Sutskever worked on Google Brain co-authored papers on TensorFlow and AlphaGo. 2015, Sutskever left Google to become confounder and chief scientist of OpenAI. In 2022, Sutskever tweeted, "it may be that today's large neural networks are slightly conscious", he is considered to have played a key role in the development of ChatGPT. Sutskever was part of the board members that fired Sam Altman in 2023 (rumoured to be about conflict over extend to which the company commit to AI safety, but he was fired for about a week only ☺) May 2024, Sutskever departure from OpenAI and started Safe Superintelligence Inc.

Alex Krizhevsky, worked at Google till 2017 and left Google after losing interest in the work.

ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky
University of Toronto
kriz@cs.utoronto.ca

Ilya Sutskever
Univ
ilya@cs.toronto.edu

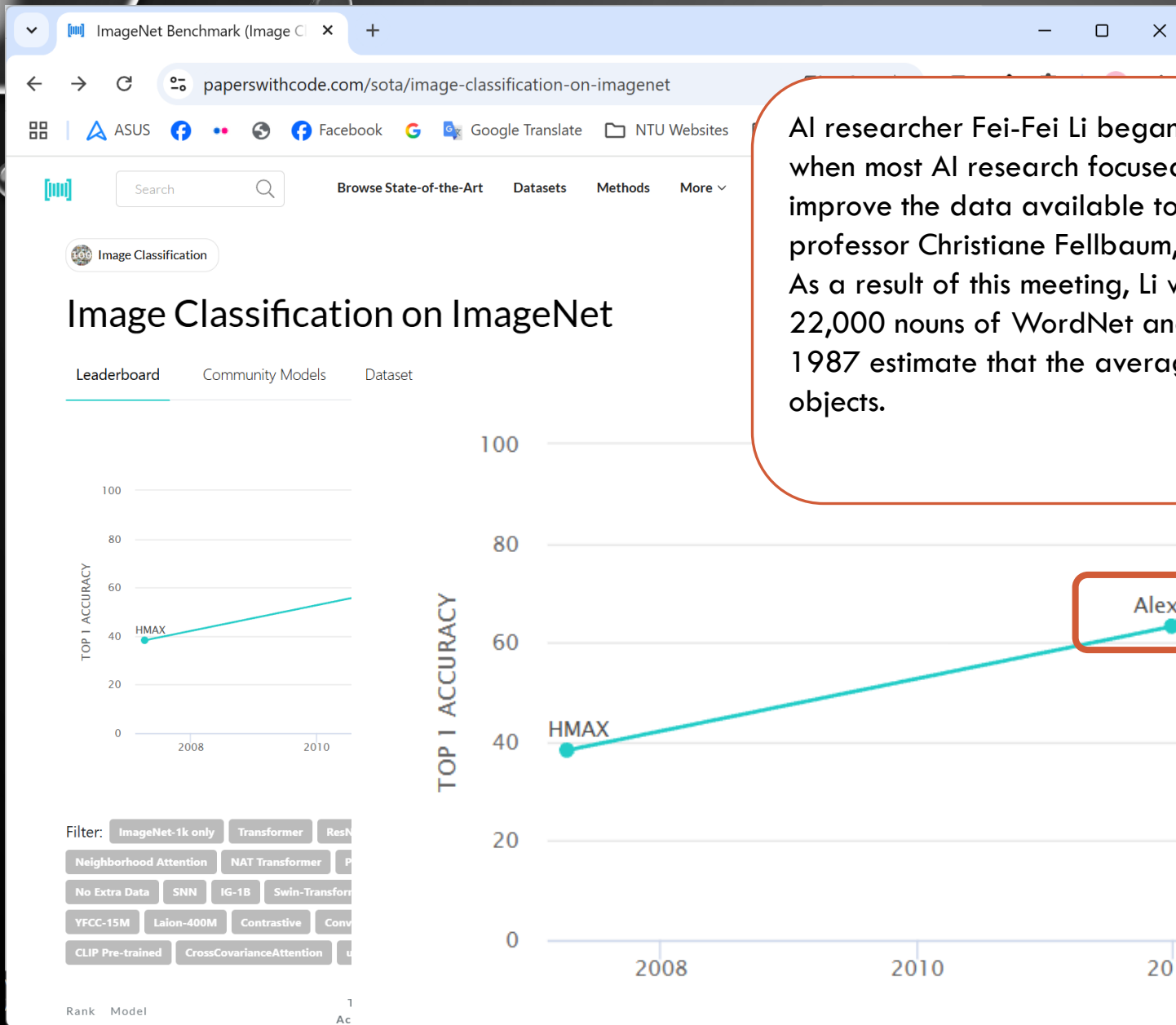
Geoffrey E. Hinton

We trained a large, deep convolutional neural network to classify high-resolution images in the ImageNet dataset.

Geoffrey Hilton known as "the Godfather of AI", 2018 Turing Award winner with 2 others for their work on deep learning, 2024 Nobel Prize for Physics. Work part-time at Google from 2013-2023, May 2023, Hinton announced his resignation from Google to be able to "freely speak out about the risks of A.I."

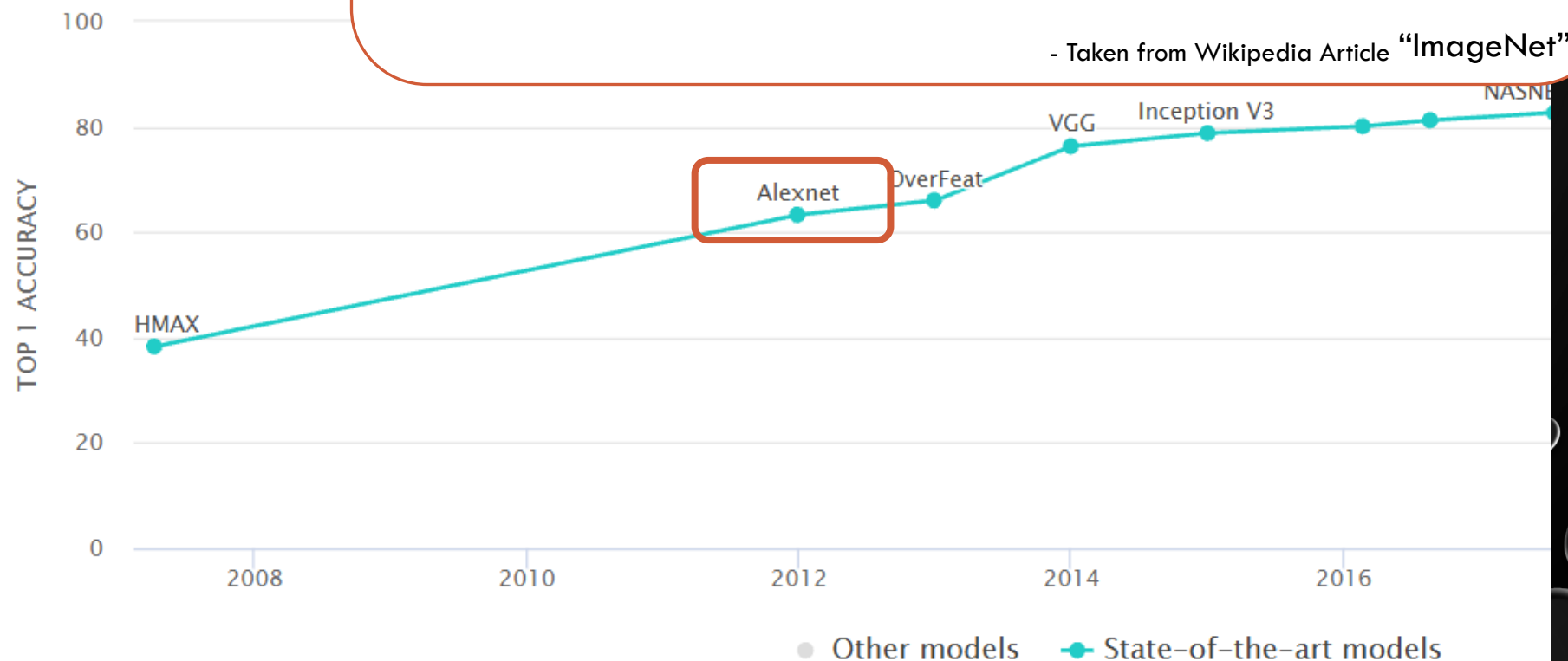
are followed by max-pooling layers, 1000-way softmax. To make training a very efficient GPU implementation and to avoid overfitting in the fully-connected layer, we used a regularization method called "dropout" [5]. We trained a variant of this model in the ImageNet challenge, winning top-5 test error rate of 15.3%, best entry.

Initial use of machine learning methods. To improve performance, we learned more powerful models, and used better datasets of labeled images were relatively small (e.g., NORB [16], Caltech-101/256 [8, 9], and CIFAR-100 [10]). These models solved quite well with datasets of this size, but they were not robust to transformations. For example, the current state-of-the-art approaches (e.g., [4]) approach human performance (<0.3%) on ImageNet, but they are highly variable, so to learn to recognize them it is necessary to use large datasets. Indeed, the shortcomings of small image datasets (e.g., [21]), but it has only recently become possible to collect labeled datasets with millions of images. The new larger datasets include LabelMe [23], which consists of hundreds of thousands of fully-segmented images, and ImageNet [6], which consists of over 15 million labeled high-resolution images in over 22,000 categories.



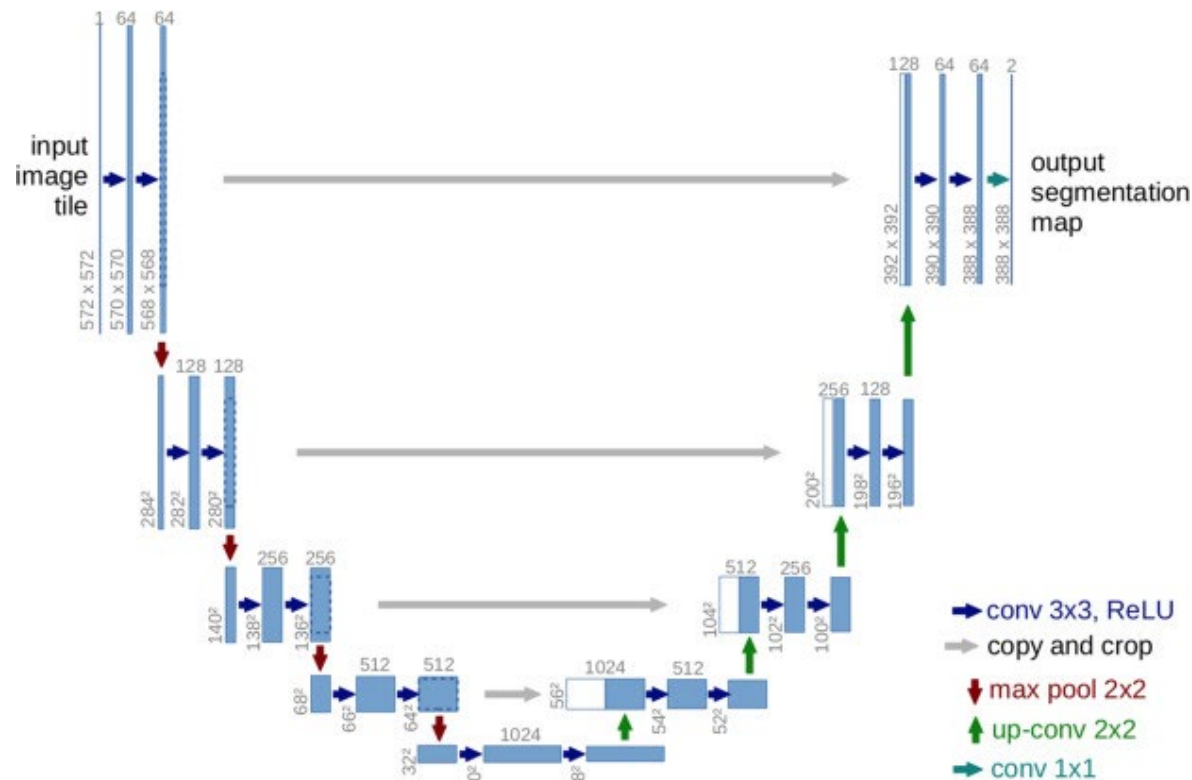
AI researcher Fei-Fei Li began working on the idea for ImageNet in 2006. At a time when most AI research focused on models and algorithms; Li wanted to expand and improve the data available to train AI algorithms. In 2007, Li met with Princeton professor Christiane Fellbaum, one of the creators of WordNet, to discuss the project. As a result of this meeting, Li went on to build ImageNet starting from the roughly 22,000 nouns of WordNet and using many of its features. She was also inspired by a 1987 estimate that the average person recognizes roughly 30,000 different kinds of objects.

- Taken from Wikipedia Article "ImageNet"



U-NET

Architecture



LATENT DIFFUSION MODELS

High-Resolution Image Synthesis with Latent Diffusion Models

Robin Rombach¹ · Andreas Blattmann¹ · Dominik Lorenz¹ Patrick Esser² Björn Ommer¹
¹Ludwig-Maximilians-University of Munich & RWTH Aachen University, Germany ²Stability AI
<https://github.com/CompVis/latent-diffusion>

Abstract

By decomposing the image formation process into a sequential application of denoising autoencoders, diffusion models (DMs) achieve state-of-the-art synthesis results on image data and beyond. Additionally, their formulation allows for a guiding mechanism to control the image generation process without retraining. However, since these models typically operate directly in pixel space, optimization of powerful DMs often consumes hundreds of GPU days and inference is expensive due to sequential evaluations. To enable DM training on limited computational resources while retaining their quality and flexibility, we apply them in the latent space of powerful pretrained autoencoders. In contrast to previous work, training diffusion models on such a representation allows for the first time to reach a near-optimal point between complexity reduction and detail preservation, greatly boosting visual fidelity. By introducing cross-attention layers into the model architecture, we turn diffusion models into powerful and flexible generators for general conditioning inputs such as text or bounding boxes and high-resolution synthesis becomes possible in a convolutional manner. Our latent diffusion models (LDMs) achieve new state-of-the-art scores for image inpainting and class-conditional image synthesis and highly competitive performance on various tasks, including text-to-image synthesis, unconditional image generation and super-resolution, while significantly reducing computational requirements compared to pixel-based DMs.

1. Introduction

Image synthesis is one of the computer vision fields with the most spectacular recent development, but also among those with the greatest computational demands. Especially high-resolution synthesis of complex, natural scenes is presently dominated by scaling up likelihood-based models, potentially containing billions of parameters in autoregressive (AR) transformers [16, 67]. In contrast, the prevailing results of GANs [3, 17, 40] have been revealed to be mostly confined to data with comparably limited variability as their adversarial learning procedure does not easily scale to modeling complex, multi-modal distributions. Recently, diffusion models [12], which are built from a hierarchy of denoising autoencoders, have shown to achieve impressive

^{*}The first two authors contributed equally to this work.

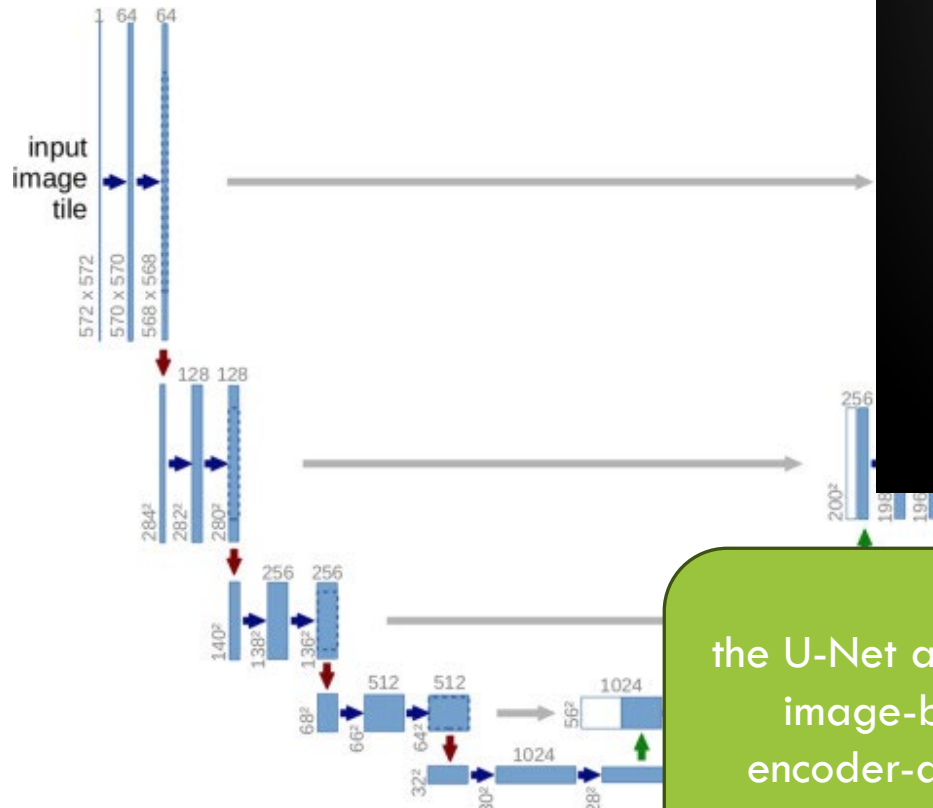


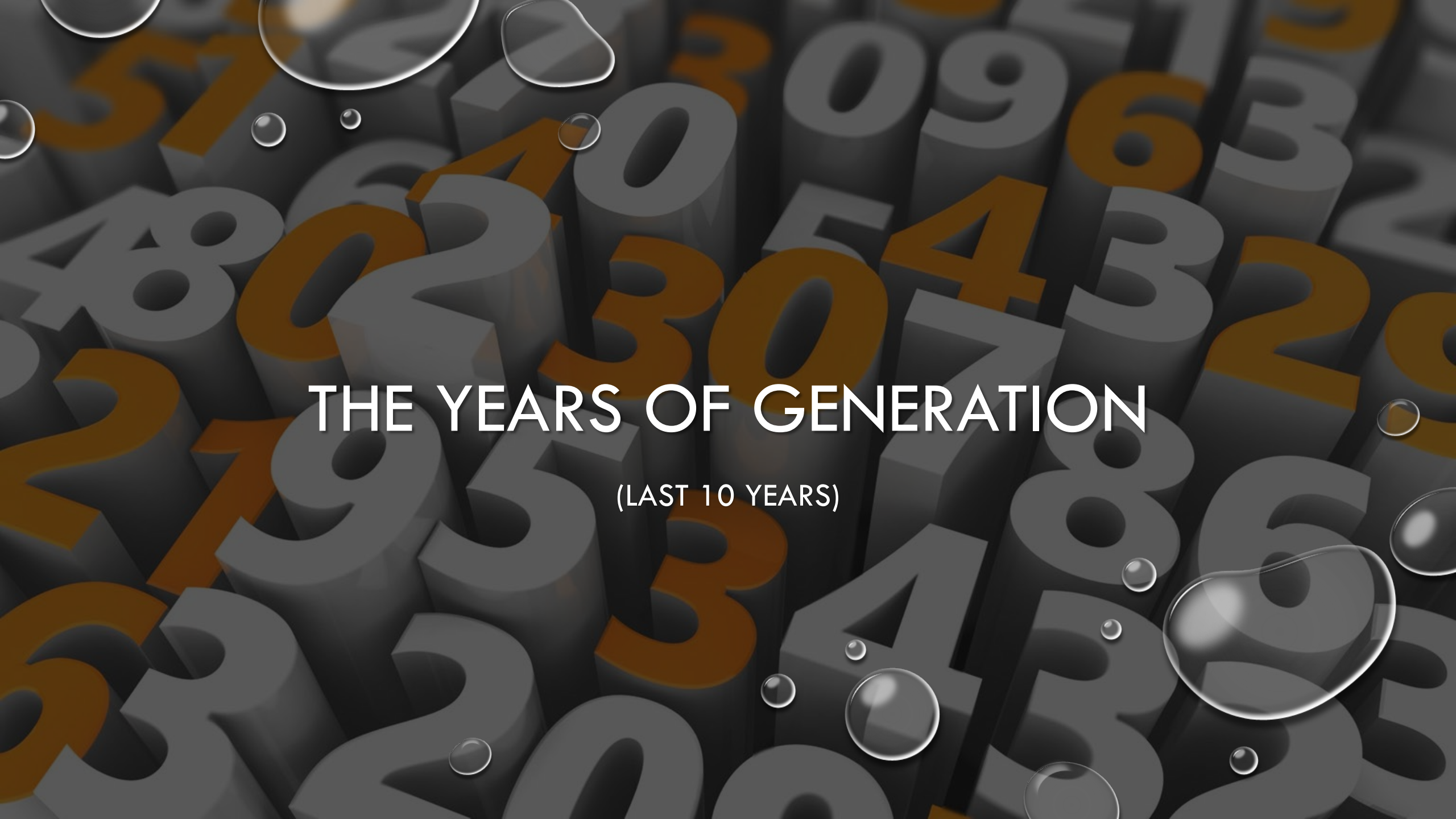
Figure 1. Boosting the upper bound on achievable quality with less aggressive downsampling. Since diffusion models offer excellent inductive biases for spatial data, we do not need the heavy spatial downsampling of related generative models in latent space, but can still greatly reduce the dimensionality of the data via suitable autoencoding models, see Sec. 3. Images are from the DDPM [1] validation set, evaluated at 512² px. We denote the spatial downsampling factor by J . Reconstruction FID [23] and PSNR are calculated on ImageNet-val [12], see also Tab. 3.

results in image synthesis [8, 11] by defining the state-of-the-art in class synthesis [15, 1] and super-resolution unconditional DMs can readily be adapted as inpainting and colorization [50] thesis [13], in contrast to other type of models [18, 46, 69]. Being likelihood-based exhibit mode-collapse and training is and, by heavily exploiting parameters model highly complex distributions of out involving billions of parameters as Democratizing High-Resolution Im- belonging to the class of likelihood-based model-covering behavior makes them excessive amounts of capacity (and thus on modeling imperceptible details of though the reweighted variational ob- dress this by undersampling the initial are still computationally demanding evaluating such a model requires repetitions (and gradient computations) in the space of RGB images. As an example, training a powerful DM often takes hundreds of GPU days (e.g. 150-1000 V100 days in [12]) and repeated evaluations on a noisy version of the input space render also inference expensive,

Stable Diffusion originated from a project called Latent Diffusion, developed in Germany by researchers at Ludwig Maximilian University in Munich and Heidelberg University. Four of the original 5 authors (Robin Rombach, Andreas Blattmann, Patrick Esser and Dominik Lorenz) later joined Stability AI and released subsequent versions of Stable Diffusion.

the U-Net architecture is frequently used in diffusion models, particularly for image-based tasks. It's a convolutional neural network (CNN) with an encoder-decoder structure and skip connections, making it well-suited for tasks like image generation and denoising.



The background of the slide is a dense, overlapping field of 3D numbers in two colors: a dark charcoal grey and a muted, earthy brown. The numbers are rendered with a strong sense of depth, casting shadows and catching highlights. Scattered across this numerical landscape are numerous water droplets of varying sizes. Some are large and clear, showing internal reflections, while others are smaller and more delicate. The overall composition is abstract and textured, suggesting themes of time, data, or the passage of years.

THE YEARS OF GENERATION

(LAST 10 YEARS)

Generative Adversarial Nets

Ian J. Goodfellow*, Jean Pouget-Abadie†, Mehdi Mirza, Bing Xu, David Warde-Farley,
Sherjil Ozair‡, Aaron Courville, Yoshua Bengio§
Département d'informatique et de recherche opérationnelle
Université de Montréal
Montréal, QC H3C 3J7

Abstract

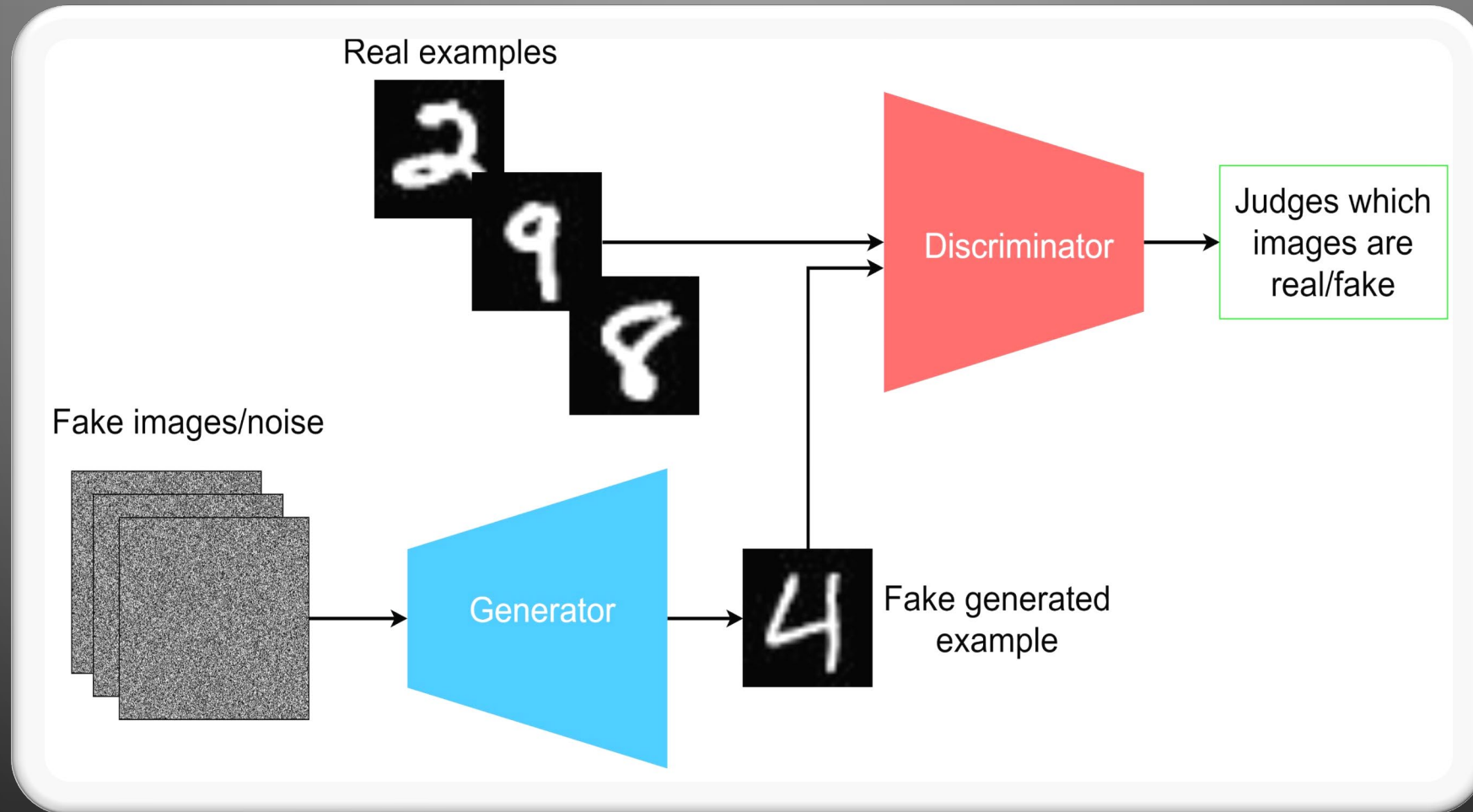
We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G . The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D , a unique solution exists, with G recovering the training data distribution and D equal to $\frac{1}{2}$ everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.

1 Introduction

The promise of deep learning is to discover rich, hierarchical models [2] that represent probability distributions over the kinds of data encountered in artificial intelligence applications, such as natural images, audio waveforms containing speech, and symbols in natural language corpora. So far, the most striking successes in deep learning have involved discriminative models, usually those that map a high-dimensional, rich sensory input to a class label [14, 20]. These striking successes have primarily been based on the backpropagation and dropout algorithms, using piecewise linear units [17, 8, 9] which have a particularly well-behaved gradient. Deep *generative* models have had less of an impact, due to the difficulty of approximating many intractable probabilistic computations that arise in maximum likelihood estimation and related strategies, and due to difficulty of leveraging the benefits of piecewise linear units in the generative context. We propose a new generative model estimation procedure that sidesteps these difficulties.¹

In the proposed *adversarial nets* framework, the generative model is pitted against an adversary: a discriminative model that learns to determine whether a sample is from the model distribution or the data distribution. The generative model can be thought of as analogous to a team of counterfeiters, trying to produce fake currency and use it without detection, while the discriminative model is analogous to the police, trying to detect the counterfeit currency. Competition in this game drives both teams to improve their methods until the counterfeits are indistinguishable from the genuine articles.

GAN
(GENERATIVE ADVERSARIAL NETS)



Analyzing and Improving the Image Quality of StyleGAN

Tero Karras
NVIDIA

Samuli Laine
NVIDIA

Miika Aittala
NVIDIA

Janne Hellsten
NVIDIA

Jaakko Lehtinen
NVIDIA and Aalto University

Timo Aila
NVIDIA

Abstract

The style-based GAN architecture (StyleGAN) yields state-of-the-art results in data-driven unconditional generative image modeling. We expose and analyze several of its characteristic artifacts, and propose changes in both model architecture and training methods to address them. In particular, we redesign the generator normalization, revisit progressive growing, and regularize the generator to encourage good conditioning in the mapping from latent codes to images. In addition to improving image quality, this path length regularizer yields the additional benefit that the generator becomes significantly easier to invert. This makes it possible to reliably attribute a generated image to a particular network. We furthermore visualize how well the generator utilizes its output resolution, and identify a capacity problem, motivating us to train larger models for additional quality improvements. Overall, our improved model redefines the state of the art in unconditional image modeling, both in terms of existing distribution quality metrics as well as perceived image quality.

1. Introduction

The resolution and quality of images produced by generative methods, especially generative adversarial networks (GAN) [16], are improving rapidly [23, 31, 5]. The current state-of-the-art method for high-resolution image synthesis is StyleGAN [24], which has been shown to work reliably on a variety of datasets. Our work focuses on fixing its characteristic artifacts and improving the result quality further.

The distinguishing feature of StyleGAN [24] is its unconventional generator architecture. Instead of feeding the input latent code $z \in \mathcal{Z}$ only to the beginning of a the network, the mapping network F first transforms it to an intermediate latent code $w \in \mathcal{W}$. Affine transforms then produce styles that control the layers of the synthesis network g via adaptive instance normalization (AdaIN) [21, 9, 13, 8]. Additionally, stochastic variation is facilitated by providing

additional random noise maps to the synthesis network. It has been demonstrated [24, 38] that this design allows the intermediate latent space $\mathcal{W}$ to be much less entangled than the input latent space $\mathcal{Z}$. In this paper, we focus all analysis solely on $\mathcal{W}$, as it is the relevant latent space from the synthesis network's point of view.

Many observers have noticed characteristic artifacts in images generated by StyleGAN [3]. We identify two causes for these artifacts, and describe changes in architecture and training methods that eliminate them. First, we investigate the origin of common blob-like artifacts, and find that the generator creates them to circumvent a design flaw in its architecture. In Section 2, we redesign the normalization used in the generator, which removes the artifacts. Second, we analyze artifacts related to progressive growing [23] that has been highly successful in stabilizing high-resolution GAN training. We propose an alternative design that achieves the same goal—training starts by focusing on low-resolution images and then progressively shifts focus to higher and higher resolutions—without changing the network topology during training. This new design also allows us to reason about the effective resolution of the generated images, which turns out to be lower than expected, motivating a capacity increase (Section 4).

Quantitative analysis of the quality of images produced using generative methods continues to be a challenging topic. Fréchet inception distance (FID) [20] measures differences in the density of two distributions in the high-dimensional feature space of an InceptionV3 classifier [39]. Precision and Recall (P&R) [36, 27] provide additional visibility by explicitly quantifying the percentage of generated images that are similar to training data and the percentage of training data that can be generated, respectively. We use these metrics to quantify the improvements.

Both FID and P&R are based on classifier networks that have recently been shown to focus on textures rather than shapes [12], and consequently, the metrics do not accurately capture all aspects of image quality. We observe that the perceptual path length (PPL) metric [24], originally introduced as a method for estimating the quality of latent space

A Style-Based Generator Architecture for Generative Adversarial Networks

Tero Karras
NVIDIA

Samuli Laine
NVIDIA

Timo Aila
NVIDIA

tkarras@nvidia.com

slaine@nvidia.com

taaila@nvidia.com

Abstract

We propose an alternative generator architecture for generative adversarial networks, borrowing from style transfer literature. The new architecture leads to an automatically learned, unsupervised separation of high-level attributes (e.g., pose and identity when trained on human faces) and stochastic variation in the generated images (e.g., freckles, hair), and it enables intuitive, scale-specific control of the synthesis. The new generator improves the state-of-the-art in terms of traditional distribution quality metrics, leads to demonstrably better interpolation properties, and also better disentangles the latent factors of variation. To quantify interpolation quality and disentanglement, we propose two new, automated methods that are applicable to any generator architecture. Finally, we introduce a new, highly varied and high-quality dataset of human faces.

1. Introduction

The resolution and quality of images produced by generative methods—especially generative adversarial networks (GAN) [22]—have seen rapid improvement recently [30, 45, 5]. Yet the generators continue to operate as black boxes, and despite recent efforts [3], the understanding of various aspects of the image synthesis process, e.g., the origin of stochastic features, is still lacking. The properties of the latent space are also poorly understood, and the commonly demonstrated latent space interpolations [13, 52, 37] provide no quantitative way to compare different generators against each other.

Motivated by style transfer literature [27], we re-design the generator architecture in a way that exposes novel ways to control the image synthesis process. Our generator starts from a learned constant input and adjusts the “style” of the image at each convolution layer based on the latent code, therefore directly controlling the strength of image features at different scales. Combined with noise injected directly into the network, this architectural change leads to automatic, unsupervised separation of high-level attributes

(e.g., pose, identity) from stochastic variation (e.g., freckles, hair) in the generated images, and enables intuitive scale-specific mixing and interpolation operations. We do not modify the discriminator or the loss function in any way, and our work is thus orthogonal to the ongoing discussion about GAN loss functions, regularization, and hyperparameters [24, 45, 5, 40, 44, 36].

Our generator embeds the input latent code into an intermediate latent space, which has a profound effect on how the factors of variation are represented in the network. The input latent space must follow the probability density of the training data, and we argue that this leads to some degree of unavoidable entanglement. Our intermediate latent space is free from that restriction and is therefore allowed to be disentangled. As previous methods for estimating the degree of latent space disentanglement are not directly applicable in our case, we propose two new automated metrics—perceptual path length and linear separability—for quantifying these aspects of the generator. Using these metrics, we show that compared to a traditional generator architecture, our generator admits a more linear, less entangled representation of different factors of variation.

Finally, we present a new dataset of human faces (Flickr-Faces-HQ, FFHQ) that offers much higher quality and covers considerably wider variation than existing high-resolution datasets (Appendix A). We have made this dataset publicly available, along with our source code and pre-trained networks.¹ The accompanying video can be found under the same link.

2. Style-based generator

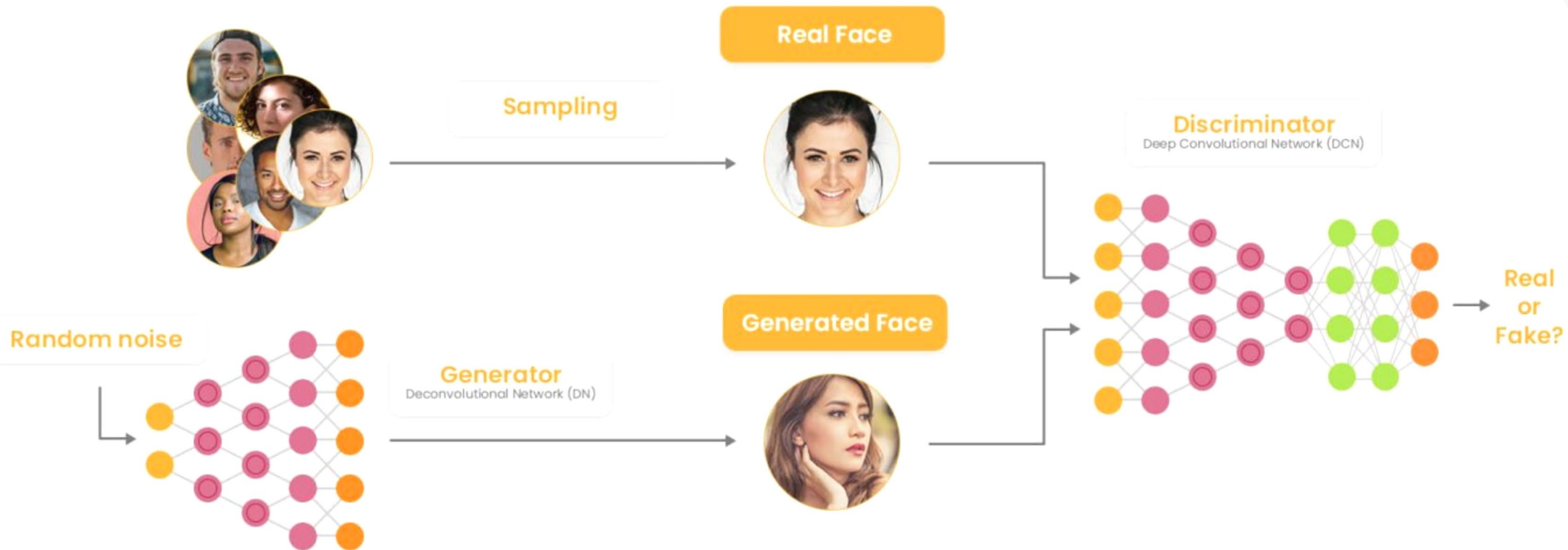
Traditionally the latent code is provided to the generator through an input layer, i.e., the first layer of a feed-forward network (Figure 1a). We depart from this design by omitting the input layer altogether and starting from a learned constant instead (Figure 1b, right). Given a latent code z in the input latent space $\mathcal{Z}$, a non-linear mapping network $f: \mathcal{Z} \rightarrow \mathcal{W}$ first produces $w \in \mathcal{W}$ (Figure 1b, left). For simplicity, we set the dimensionality of both

¹<https://github.com/NVlabs/stylegan>

StyleGAN

In December 2018, Nvidia researchers distributed a preprint with accompanying software introducing StyleGAN, a GAN for producing an unlimited number of (often convincing) portraits of fake human faces. StyleGAN was able to run on Nvidia's commodity GPU processors

In February 2019, Uber engineer Phillip Wang used the software to create the website This Person Does Not Exist, which displayed a new face on each web page reload.



HAVE SOME FUN

[HTTPS://THIS-PERSON-DOES-NOT-EXIST.COM/EN](https://this-person-does-not-exist.com/en)

Random Face Generator (This Person Does Not Exist)

Generate random human face in 1 click and download it! AI generated fake person photos: man, woman or child.

Gender:

Any



Age:

Any



Ethnicity:

Any



Refresh Image



Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez* †
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukaszkaiser@google.com

Illia Polosukhin* ‡
illia.polosukhin@gmail.com

Abstract

The modern version of the transformer was proposed in the 2017 paper "Attention Is All You Need" by researchers at Google. They are used in large-scale natural language processing, computer vision (vision transformers), reinforcement learning, audio, multimodal learning, robotics, and even playing chess. It has also led to the development of pre-trained systems, such as generative pre-trained transformers (GPTs) and BERT[12] (bidirectional encoder representations from transformers).

the
has been
attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research

TRANSFORMER

(deep learning architecture)

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez*[†]
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukaszkaiser@google.com

Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Łukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

TRANSFORMER

(deep learning architecture)

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Alexey Dosovitskiy^{*,†}, Lucas Beyer^{*}, Alexander Kolesnikov^{*}, Dirk Weissenborn^{*},
Xiaohua Zhai^{*}, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer,
Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby^{*,†}

^{*}equal technical contribution, [†]equal advising

Google Research, Brain Team

{adosovitskiy, neilhoulby}@google.com

ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.¹

1 INTRODUCTION

Self-attention-based architectures, in particular Transformers (Vaswani et al., 2017), have become the model of choice in natural language processing (NLP). The dominant approach is to pre-train on a large text corpus and then fine-tune on a smaller task-specific dataset (Devlin et al., 2019). Thanks to Transformers’ computational efficiency and scalability, it has become possible to train models of unprecedented size, with over 100B parameters (Brown et al., 2020; Lepikhin et al., 2020). With the models and datasets growing, there is still no sign of saturating performance.

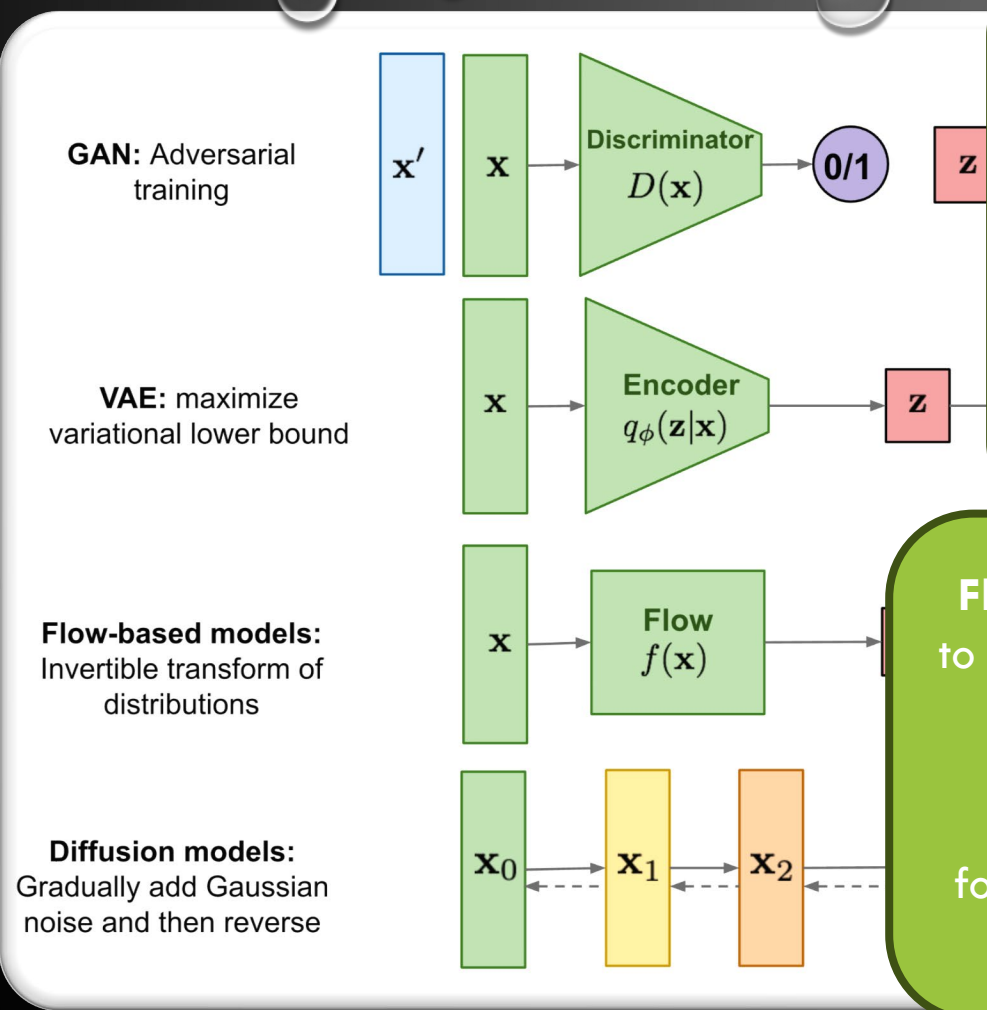
In computer vision, however, convolutional architectures remain dominant (LeCun et al., 1989; Krizhevsky et al., 2012; He et al., 2016). Inspired by NLP successes, multiple works try combining CNN-like architectures with self-attention (Wang et al., 2018; Carion et al., 2020), some replacing the convolutions entirely (Ramachandran et al., 2019; Wang et al., 2020a). The latter models, while theoretically efficient, have not yet been scaled effectively on modern hardware accelerators due to the use of specialized attention patterns. Therefore, in large-scale image recognition, classic ResNet-like architectures are still state of the art (Mahajan et al., 2018; Xie et al., 2020; Kolesnikov et al., 2020).

Inspired by the Transformer scaling successes in NLP, we experiment with applying a standard Transformer directly to images, with the fewest possible modifications. To do so, we split an image into patches and provide the sequence of linear embeddings of these patches as an input to a Transformer. Image patches are treated the same way as tokens (words) in an NLP application. We train the model on image classification in supervised fashion.

When trained on mid-sized datasets such as ImageNet without strong regularization, these models yield modest accuracies of a few percentage points below ResNets of comparable size. This seemingly discouraging outcome may be expected: Transformers lack some of the inductive biases

ViT

Vision Transformer



In a **GAN**, two neural networks compete with each other in the form of a zero-sum game, where one agent's gain is another agent's loss.

In **VAE**, (variational autoencoder) The first neural network (encoder) takes as input the data points themselves, and outputs parameters for the variational distribution. As it maps from a known input space to the low-dimensional latent space. The second neural network of this model (decoder) maps from the latent space to the input space, e.g. as the means of the noise distribution.

Flow-based models use a sequence of inverse transformations to generate the image unlike other flow-based models and learn the probability distribution.

Diffusion models will add noise to the data samples in the forward process and in the reverse process they will reconstruct the data samples from the noise.

Deep Unsupervised Learning using Nonequilibrium Thermodynamics

Jascha Sohl-Dickstein
Stanford University

Eric A. Weiss
University of California, Berkeley

Niru Maheswaranathan
Stanford University

Surya Ganguli
Stanford University

JASCHA@STANFORD.EDU

EAWISS@BERKELEY.EDU

NIRUM@STANFORD.EDU

SGANGULI@STANFORD.EDU

Abstract

A central problem in machine learning involves modeling complex data-sets using highly flexible families of probability distributions in which learning, sampling, inference, and evaluation are still analytically or computationally tractable. Here, we develop an approach that simultaneously achieves both flexibility and tractability. The essential idea, inspired by non-equilibrium statistical physics, is to systematically and slowly destroy structure in a data distribution through an iterative forward diffusion process. We then learn a reverse diffusion process that restores structure in data, yielding a highly flexible and tractable generative model of the data. This approach allows us to rapidly learn, sample from, and evaluate probabilities in deep generative models with thousands of layers or time steps, as well as to compute conditional and posterior probabilities under the learned model. We additionally release an open source reference implementation of the algorithm.

1. Introduction

Historically, probabilistic models suffer from a tradeoff between two conflicting objectives: *tractability* and *flexibility*. Models that are *tractable* can be analytically evaluated and easily fit to data (e.g. a Gaussian or Laplace). However,

Proceedings of the 32nd International Conference on Machine Learning, Lille, France, 2015. JMLR: W&CP volume 37. Copyright 2015 by the author(s).

these models are unable to aptly describe structure in rich datasets. On the other hand, models that are *flexible* can be molded to fit structure in arbitrary data. For example, we can define models in terms of any (non-negative) function $\phi(\mathbf{x})$ yielding the flexible distribution $p(\mathbf{x}) = \frac{\phi(\mathbf{x})}{Z}$, where Z is a normalization constant. However, computing this normalization constant is generally intractable. Evaluating, training, or drawing samples from such flexible models typically requires a very expensive Monte Carlo process.

A variety of analytic approximations exist which ameliorate, but do not remove, this tradeoff—for instance mean field theory and its expansions (T, 1982; Tanaka, 1998), variational Bayes (Jordan et al., 1999), contrastive divergence (Welling & Hinton, 2002; Hinton, 2002), minimum probability flow (Sohl-Dickstein et al., 2011b;a), minimum KL contraction (Lyu, 2011), proper scoring rules (Gneiting & Raftery, 2007; Parry et al., 2012), score matching (Hyvärinen, 2005), pseudolikelihood (Besag, 1975), loopy belief propagation (Murphy et al., 1999), and many, many more. Non-parametric methods (Gershman & Blei, 2012) can also be very effective¹.

1.1. Diffusion probabilistic models

We present a novel way to define probabilistic models that allows:

1. extreme flexibility in model structure,
2. exact sampling,

¹Non-parametric methods can be seen as transitioning smoothly between tractable and flexible models. For instance, a non-parametric Gaussian mixture model will represent a small amount of data using a single Gaussian, but may represent infinite data as a mixture of an infinite number of Gaussians.

DIFFUSION PROBABILISTIC MODELS

DIFFUSION MODEL

2015

Deep Unsupervised Learning using Nonequilibrium Thermodynamics

Jascha Sohl-Dickstein
Stanford University

JASCHA@STANFORD.EDU

Eric A. Weiss
University of California, Berkeley

EAWEISS@BERKELEY.EDU

Niru Maheswaranathan
Stanford University

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Stanford University

SGANGULI@STANFORD.EDU

Abstract

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A variety of analytic approximations exist which ameliorate, but do not remove, this tradeoff—for instance mean field theory and its expansions (T, 1982; Tanaka, 1998), variational Bayes (Jordan et al., 1999), contrastive divergence (Welling & Hinton, 2002; Hinton, 2002), minimum probability flow (Sohl-Dickstein et al., 2011b), minimum KL contraction (Lyu, 2011), proper scoring rules (Gneiting & Raftery, 2007; Parry et al., 2012), score matching (Hyvärinen, 2005), pseudolikelihood (Besag, 1975), loopy belief propagation (Murphy et al., 1999), and many, many more. Non-parametric methods (Gershman & Blei, 2012) can also be very effective¹.

1.1. Diffusion probabilistic models

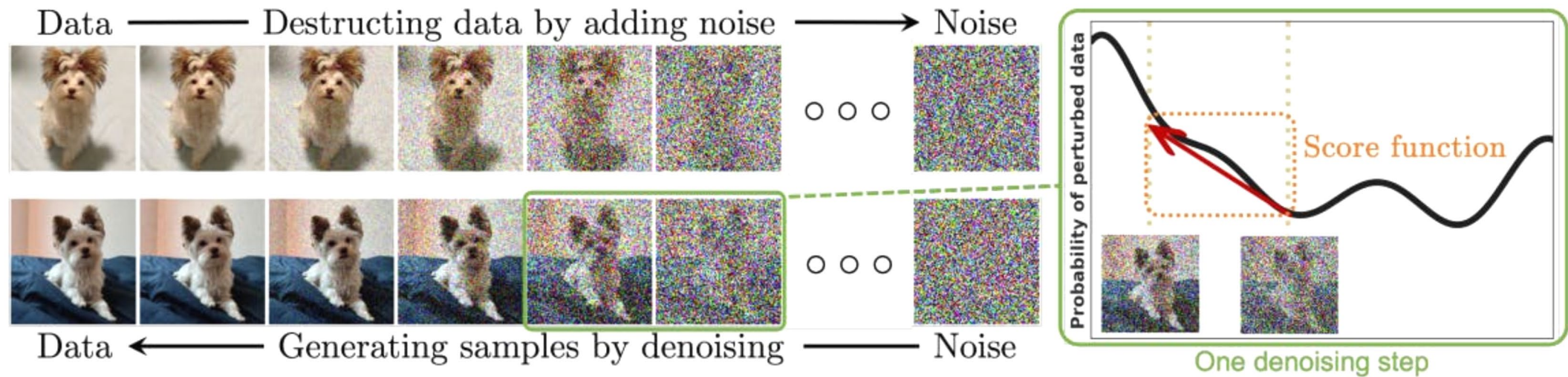
We present a novel way to define probabilistic models that allows:

1. extreme flexibility in model structure,
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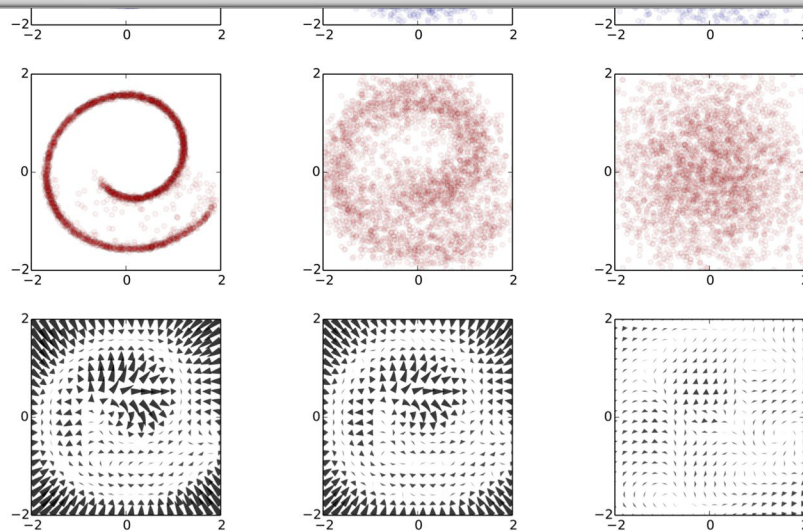
"The essential idea, inspired by non-equilibrium statistical physics, is to systematically and slowly destroy structure in a data distribution through an iterative forward diffusion process. We then learn a reverse diffusion process that restores structure in data, yielding a highly flexible and tractable generative model of the data."

Author's description in the 2015 paper "Deep Unsupervised Learning using Nonequilibrium Thermodynamics"



The reverse trajectory

$$p_{\theta}(\mathbf{x}_{0:T})$$



The drifting term

$$\mu_{\theta}(\mathbf{x}_t, t) - \mathbf{x}_t$$

The "drift" in diffusion models refers to the systematic changes introduced by the forward process, while the "noise predictor" is a neural network that learns to estimate and remove the added noise during the reverse process.

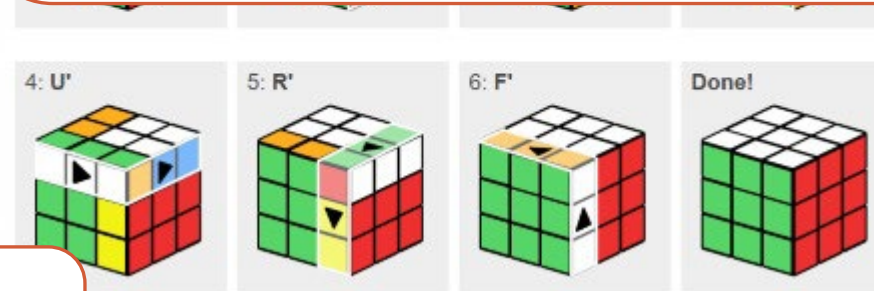
An example of training a diffusion model for modeling a 2D swiss roll data. (Image source: [Sohl-Dickstein et al., 2015](#))



Analogy : To solve the Rubik's Cube, you can learn from the available techniques/algorithms.

Diffusion method learn the noise predictor (usually via a neural network) to remove the noise gradually in each step of the reverse process

Diffusion achieve an output that is relatively close to the original (don't need to be exact replica)



LAYMAN'S DEFINITION/ANALOGY – **RUBIC'S CUBE**

High-Resolution Image Synthesis with Latent Diffusion Models

Robin Rombach¹ * Andreas Blattmann¹ * Dominik Lorenz¹ Patrick Esser^{1B} Björn Ommer¹

¹Ludwig Maximilian University of Munich & IWR, Heidelberg University, Germany ^BRunway ML

<https://github.com/CompVis/latent-diffusion>

Abstract

By decomposing the image formation process into a sequential application of denoising autoencoders, diffusion models (DMs) achieve state-of-the-art synthesis results on image data and beyond. Additionally, their formulation allows for a guiding mechanism to control the image generation process without retraining. However, since these models typically operate directly in pixel space, optimization of powerful DMs often consumes hundreds of GPU days and inference is expensive due to sequential evaluations. To enable DM training on limited computational resources while retaining their quality and flexibility, we apply them in the latent space of powerful pretrained autoencoders. In contrast to previous work, training diffusion models on such a representation allows for the first time to reach a near-optimal point between complexity reduction and detail preservation, greatly boosting visual fidelity. By introducing cross-attention layers into the model architecture, we turn diffusion models into powerful and flexible generators for general conditioning inputs such as text or bounding boxes and high-resolution synthesis becomes possible in a convolutional manner. Our latent diffusion models (LDMs) achieve new state-of-the-art scores for image inpainting and class-conditional image synthesis and highly competitive performance on various tasks, including text-to-image synthesis, unconditional image generation and super-resolution, while significantly reducing computational requirements compared to pixel-based DMs.

1. Introduction

Image synthesis is one of the computer vision fields with the most spectacular recent development, but also among those with the greatest computational demands. Especially high-resolution synthesis of complex, natural scenes is presently dominated by scaling up likelihood-based models, potentially containing billions of parameters in autoregressive (AR) transformers [66,67]. In contrast, the promising results of GANs [3, 27, 40] have been revealed to be mostly confined to data with comparably limited variability as their adversarial learning procedure does not easily scale to modeling complex, multi-modal distributions. Recently, diffusion models [82], which are built from a hierarchy of denoising autoencoders, have shown to achieve impressive



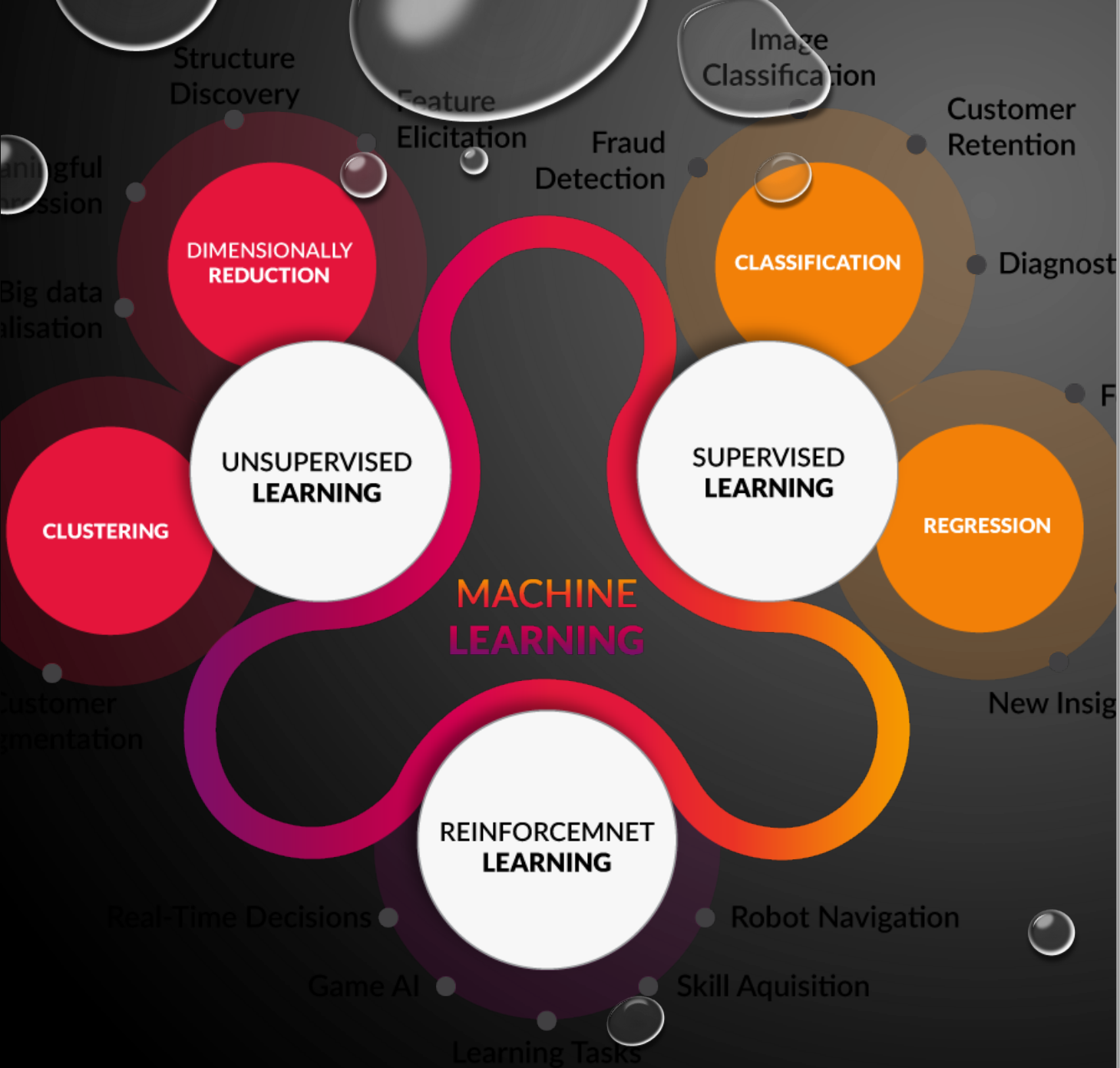
Figure 1. Boosting the upper bound on achievable quality with less aggressive downsampling. Since diffusion models offer excellent inductive biases for spatial data, we do not need the heavy spatial downsampling of related generative models in latent space, but can still greatly reduce the dimensionality of the data via suitable autoencoding models, see Sec. 3. Images are from the DIV2K [1] validation set, evaluated at 512² px. We denote the spatial downsampling factor by f . Reconstruction FIDs [29] and PSNR are calculated on ImageNet-val. [12]; see also Tab. 8.

results in image synthesis [30,85] and have recently become the state-of-the-art in class-conditional image synthesis [15,31] and super-resolution [85]. Unconditional DMs can readily be adapted to tasks such as inpainting and colorization [85] and text-to-image synthesis [53], in contrast to other types of generative models [19,46,69]. Being likelihood-based models, they exhibit mode-collapse and training instability, and, by heavily exploiting parameterized distributions, model highly complex distributions of natural images without involving billions of parameters as autoregressive models. **Democratizing High-Resolution Image Synthesis** Our models belong to the class of likelihood-based models, but their mode-covering behavior makes them more efficient in terms of capacity (and thus inference time) on modeling imperceptible details of natural images, though the reweighted variational objective function used to address this by undersampling the initial noise is still computationally demanding. Evaluating such a model requires repeated sampling (and gradient computations) in the latent space of RGB images. As an example, training our most powerful DMs often takes hundreds of GPU days (e.g. 150 - 1000 V100 days in [15]) and repeated evaluations on a noisy version of the input space render also inference expensive,

LATENT DIFFUSION MODELS

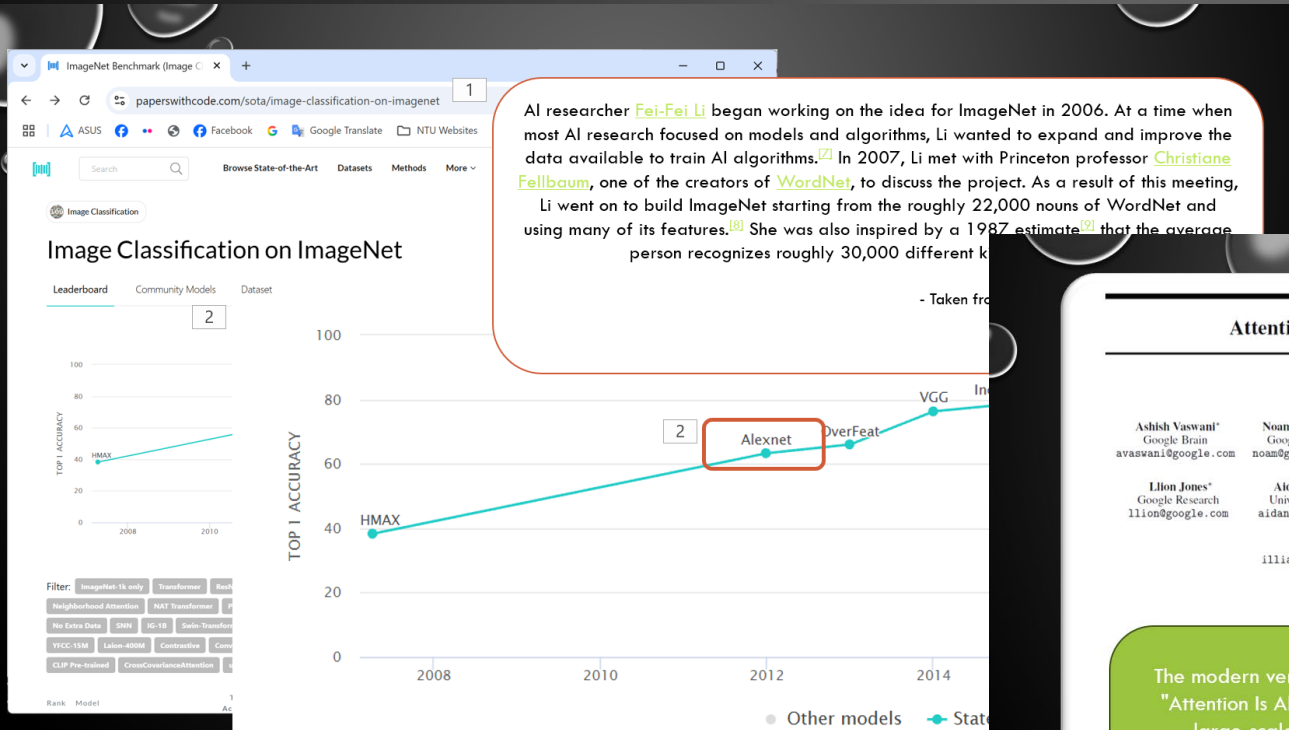
Stable Diffusion originated from a project called Latent Diffusion, developed in Germany by researchers at Ludwig Maximilian University in Munich and Heidelberg University. Four of the original 5 authors (Robin Rombach, Andreas Blattmann, Patrick Esser and Dominik Lorenz) later joined Stability AI and released subsequent versions of Stable Diffusion.

*The first two authors contributed equally to this work.



TYPICAL CLASSIFICATIONS

BREAKTHROUGH ?



AI researcher Fei-Fei Li began working on the idea for ImageNet in 2006. At a time when most AI research focused on models and algorithms, Li wanted to expand and improve the data available to train AI algorithms.^[2] In 2007, Li met with Princeton professor Christiane Fellbaum, one of the creators of WordNet, to discuss the project. As a result of this meeting, Li went on to build ImageNet starting from the roughly 22,000 nouns of WordNet and using many of its features.^[8] She was also inspired by a 1987 estimate^[9] that the average person recognizes roughly 30,000 different k

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
uszko@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez*[†]
University of Toronto
aidan@cs.toronto.edu

Lukasz Kaiser*
Google Brain
lukaszkaiser@google.com

Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

Abstract

The modern version of the transformer was proposed in the 2017 paper "Attention Is All You Need" by researchers at Google. They are used in large-scale natural language processing, computer vision (vision transformers), reinforcement learning, audio, multimodal learning, robotics, and even playing chess. It has also led to the development of pre-trained systems, such as generative pre-trained transformers (GPTs) and BERT[12] (bidirectional encoder representations from transformers).

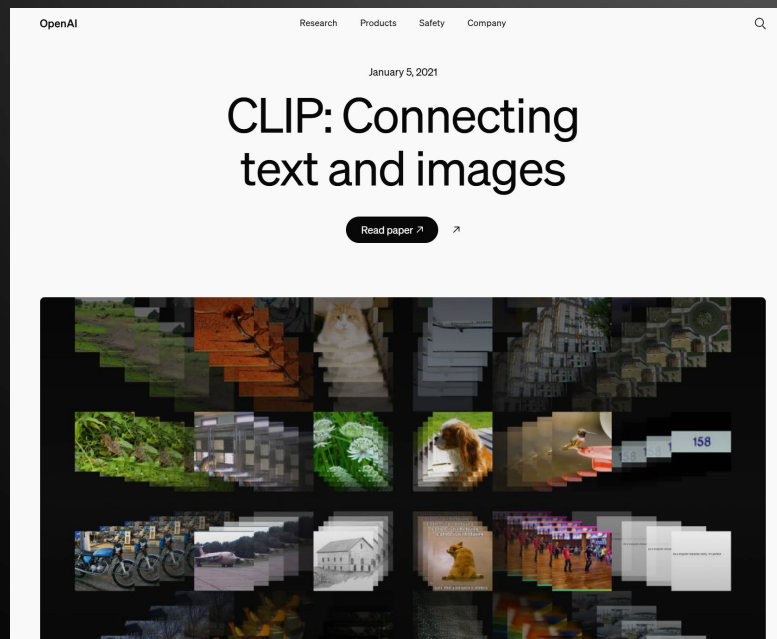
TRANSFORMER

(deep learning architecture)

2017

CLIP

(Contrastive Language-Image Pre-Training)



Learning Transferable Visual Models From Natural Language Supervision

Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹

Abstract

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study the performance of this approach by benchmarking on over 30 different existing computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we match the accuracy of the original ResNet-50 on ImageNet zero-shot without needing to use any of the 1.28 million training examples it was trained on. We release our code and pre-trained model weights at <https://github.com/OpenAI/CLIP>.

1. Introduction and Motivating Work

Pre-training methods which learn directly from raw text have revolutionized NLP over the last few years (Dai & Le, 2015; Peters et al., 2018; Howard & Ruder, 2018; Radford et al., 2018; Devlin et al., 2018; Raffel et al., 2019).

^{*}Equal contribution ¹OpenAI, San Francisco, CA 94110, USA. Correspondence to: <{alec, jongwook}@openai.com>.

Task-agnostic objectives such as autoregressive and masked language modeling have scaled across many orders of magnitude in compute, model capacity, and data, steadily improving capabilities. The development of “text-to-text” as a standardized input-output interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets removing the need for specialized output heads or dataset specific customization. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

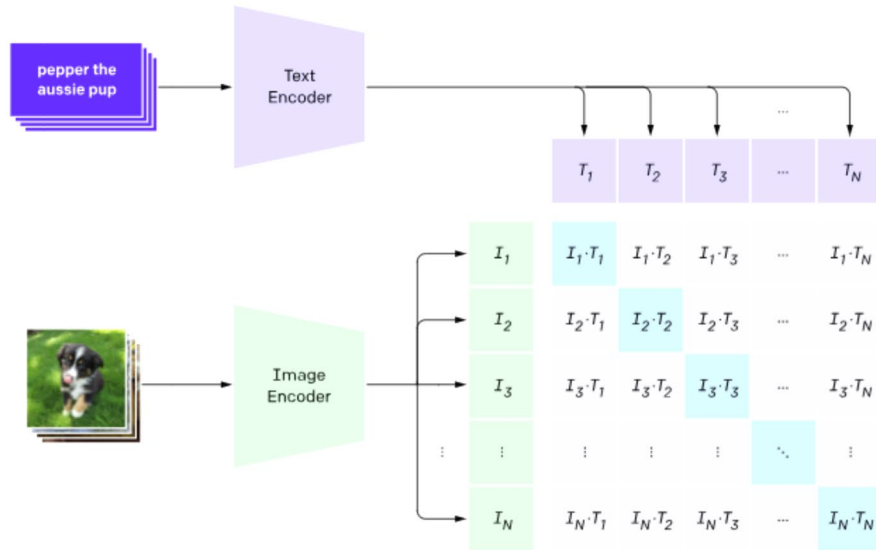
These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009). Could scalable pre-training methods which learn directly from web text result in a similar breakthrough in computer vision? Prior work is encouraging.

Over 20 years ago Mori et al. (1999) explored improving content based image retrieval by training a model to predict the nouns and adjectives in text documents paired with images. Quattoni et al. (2007) demonstrated it was possible to learn more data efficient image representations via manifold learning in the weight space of classifiers trained to predict words in captions associated with images. Srivastava & Salakhutdinov (2012) explored deep representation learning by training multimodal Deep Boltzmann Machines on top of low-level image and text tag features. Joulin et al. (2016) modernized this line of work and demonstrated that CNNs trained to predict words in image captions learn useful image representations. They converted the title, description, and hashtag metadata of images in the YFCC100M dataset (Thomee et al., 2016) into a bag-of-words multi-label classification task and showed that pre-training AlexNet (Krizhevsky et al., 2012) to predict these labels learned representations which performed similarly to ImageNet-based pre-training on transfer tasks. Li et al. (2017) then extended this approach to predicting phrase n-grams in addition to individual words and demonstrated the ability of their system to zero-shot transfer to other image

arXiv:2103.00020v1 [cs.CV] 26 Feb 2021

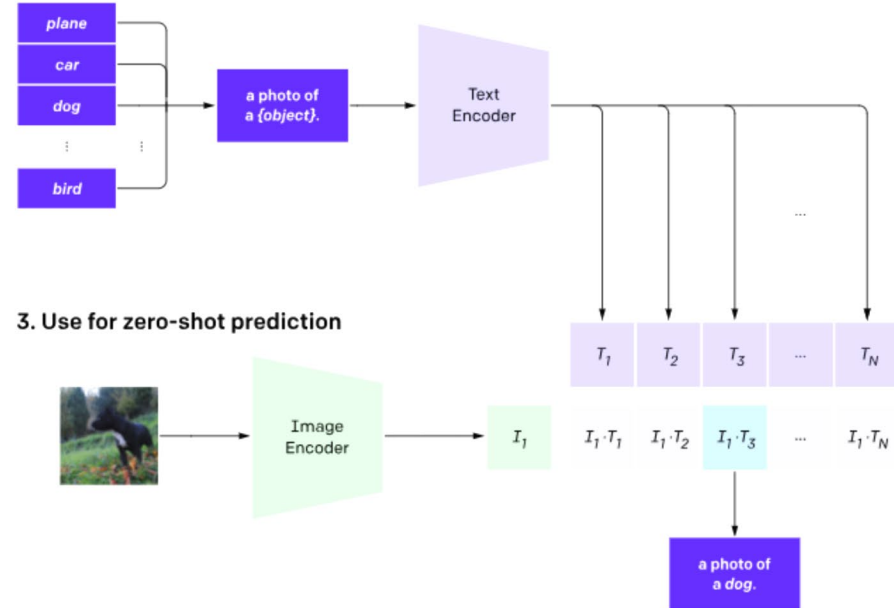
CLIP (Contrastive Language-Image Pre-Training)

1. Contrastive pre-training

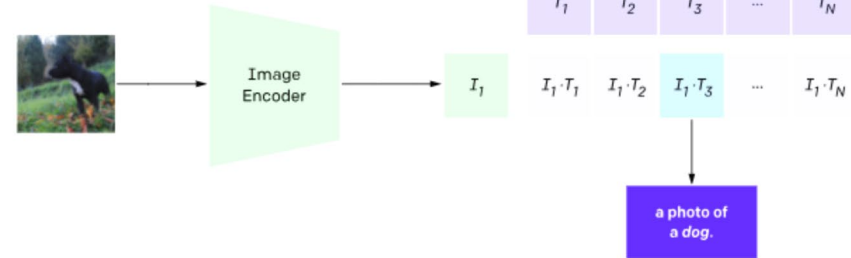


Overview A

2. Create dataset classifier from label text

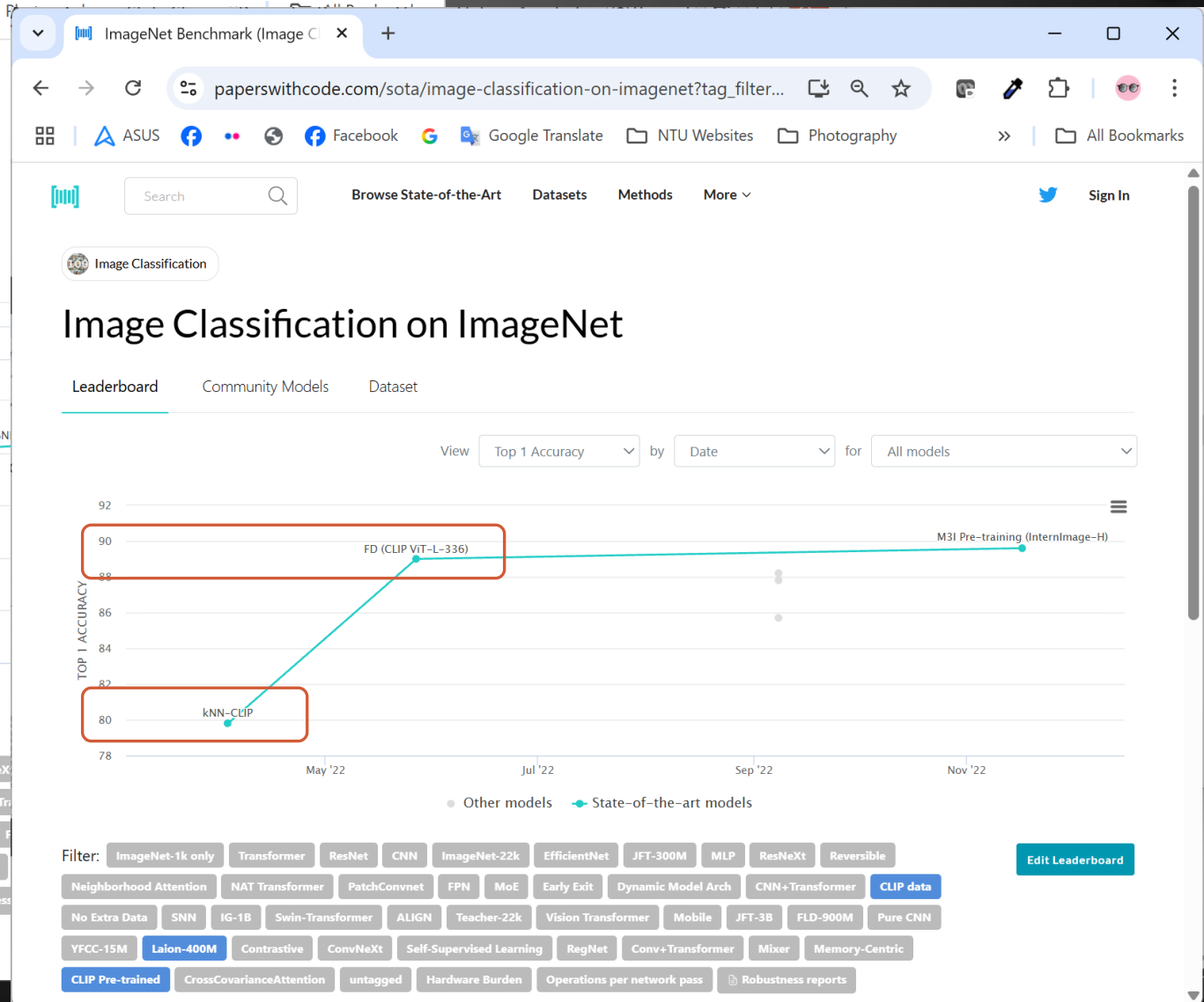
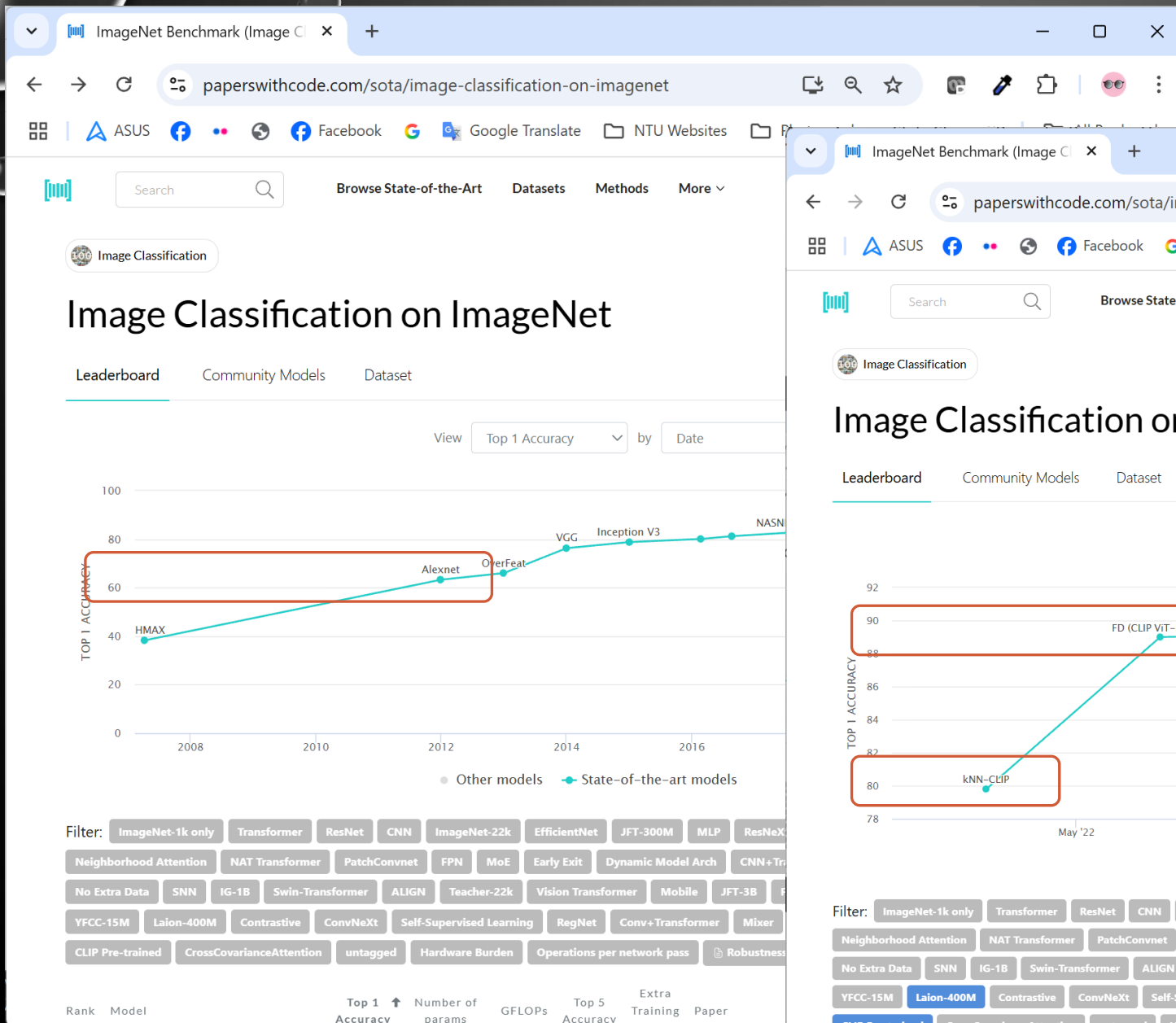


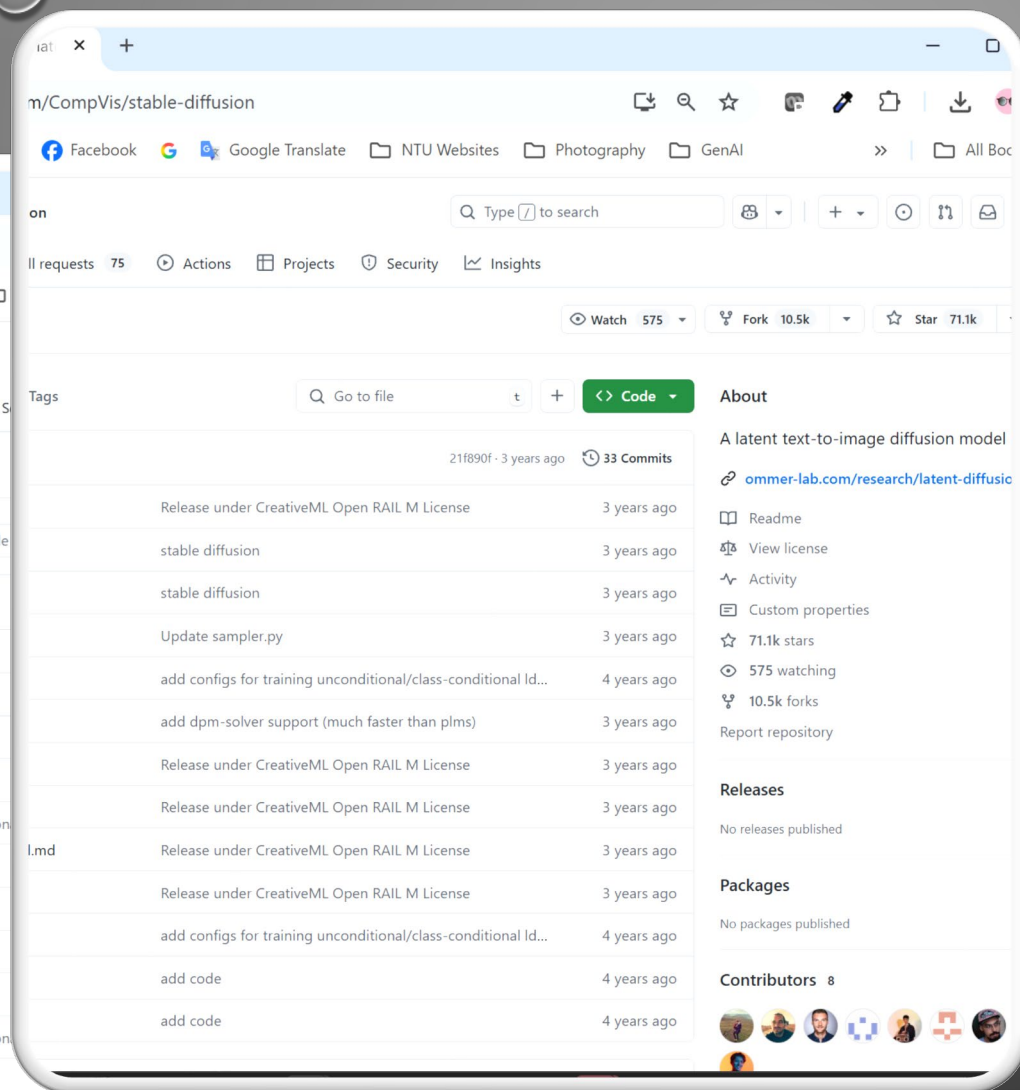
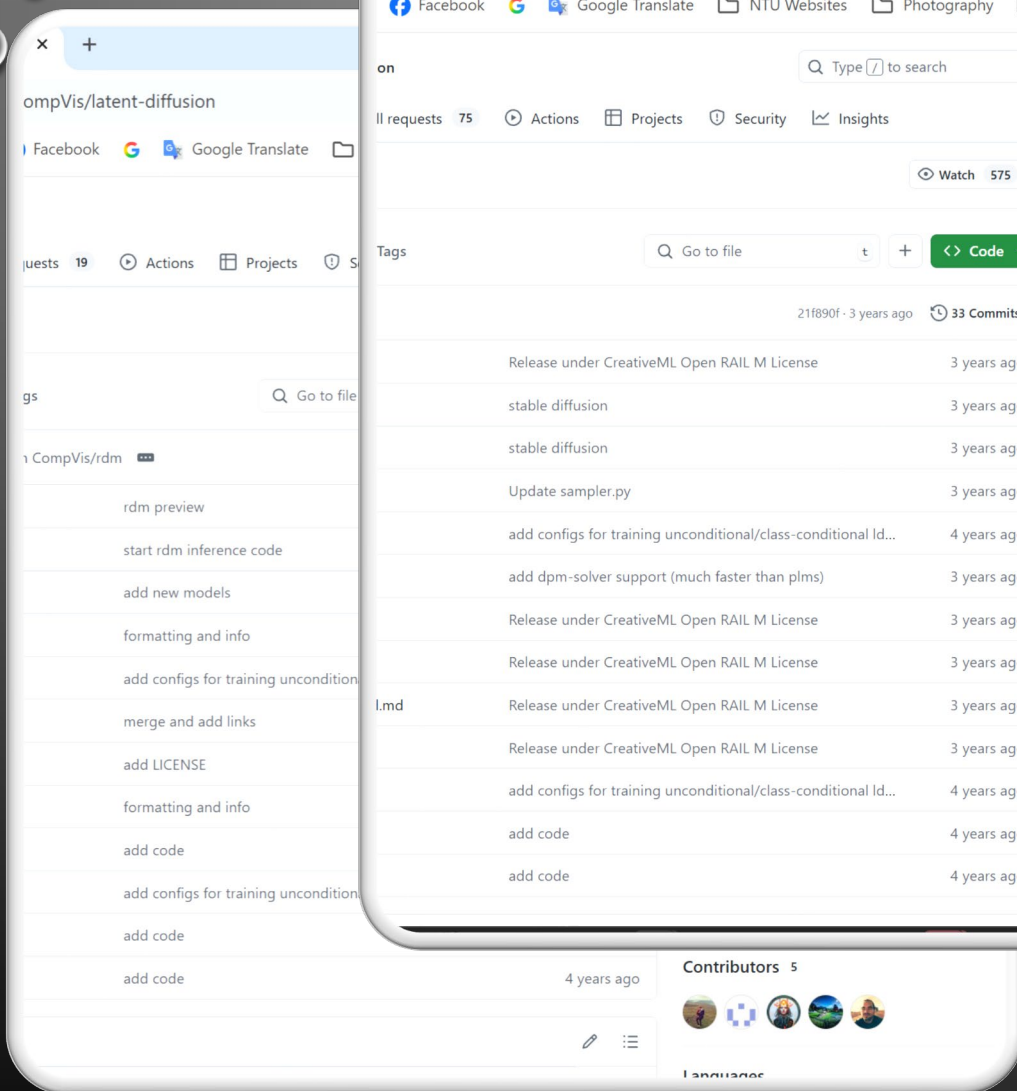
3. Use for zero-shot prediction



Overview B

CLIP pre-trains an image encoder and a text encoder to predict which images were paired with which texts in our dataset. We then use this behavior to turn CLIP into a zero-shot classifier. We convert all of a dataset's classes into captions such as "a photo of a dog" and predict the class of the caption CLIP estimates best pairs with a given image.





<https://github.com/CompVis/stable-diffusion>

A black and white photograph of a classical building facade, likely a museum or government building. The image features several statues on the roofline, including a prominent figure holding scales of justice. The building's architecture is highly detailed with ornate carvings and columns. The sky is dark and cloudy. Numerous water droplets of various sizes are scattered across the image, some in the foreground and others near the building, creating a sense of freshness or perhaps a metaphorical 'cleansing' of history.

THE REST IS HISTORY

The background features a dense, overlapping field of 3D numbers in two colors: a muted grey and a mustard yellow. The numbers are rendered with soft shadows, giving them a sense of depth and volume. Scattered throughout the scene are several translucent, realistic-looking bubbles of varying sizes, some with highlights and reflections, adding a dynamic and organic feel to the composition.

PUTTING IT TOGETHER

(LAST 10 YEARS)

STABLE DIFFUSION

STABLE DIFFUSION ORIGINATED FROM A PROJECT CALLED LATENT DIFFUSION, DEVELOPED IN GERMANY BY RESEARCHERS AT LUDWIG MAXIMILIAN UNIVERSITY IN MUNICH AND HEIDELBERG UNIVERSITY. FOUR OF THE ORIGINAL 5 AUTHORS (ROBIN ROMBACH, ANDREAS BLATTMANN, PATRICK ESSER AND DOMINIK LORENZ) LATER JOINED STABILITY AI AND RELEASED SUBSEQUENT VERSIONS OF STABLE DIFFUSION.

THE TECHNICAL LICENSE FOR THE MODEL WAS RELEASED BY THE COMPVIS GROUP AT LUDWIG MAXIMILIAN UNIVERSITY OF MUNICH. DEVELOPMENT WAS LED BY PATRICK ESSER OF RUNWAY AND ROBIN ROMBACH OF COMPVIS, WHO WERE AMONG THE RESEARCHERS WHO HAD EARLIER INVENTED THE LATENT DIFFUSION MODEL ARCHITECTURE USED BY STABLE DIFFUSION. STABILITY AI ALSO CREDITED ELEUTHERAI AND LAION (A GERMAN NONPROFIT WHICH ASSEMBLED THE DATASET ON WHICH STABLE DIFFUSION WAS TRAINED) AS SUPPORTERS OF THE PROJECT.



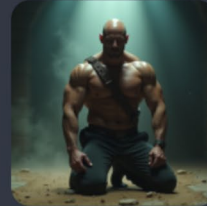


How does Stable Diffusion work?

Updated June 9, 2024 · By [Andrew](#) · Categorized as [Tutorial](#) · Tagged [Fundamentals](#), [Img2img](#), [Txt2img](#) · [45 Comments](#)

Stable Diffusion is a latent diffusion model that generates AI images from text. Instead of operating in the high-dimensional image space, it first compresses the image into the latent space.

Latest Articles



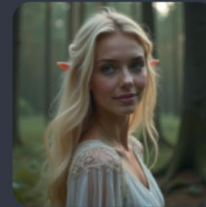
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[Cutting a gold bar](#)



[Clone Your Voice Using AI \(ComfyUI\)](#)



[How to run Wan VACE video-to-video in ComfyUI](#)

Table of Contents

What can Stable Diffusion do?

Diffusion model

Forward diffusion

Reverse diffusion

How training is done

Reverse diffusion

Stable Diffusion model

Latent diffusion model

Variational Autoencoder

Image resolution

Image upscaling

Why is latent space possible?

Reverse diffusion in latent space

What is a VAE file?

Conditioning

Text conditioning (text-to-image)

Other conditionings

Stable Diffusion step-by-step

Text-to-image

Noise schedule

Image-to-image

Inpainting

Depth-to-image

What is CFG value?

Classifier guidance

Classifier-free guidance

Stable Diffusion v1.5 vs v2

Model difference

Training data difference

Outcome difference

SDXL model

Ethical, moral and legal issues

Some interesting reads

STABLE DIFFUSION

×

Announcements

Information

Content

Assignments

Discussion Board

Online Quiz

Course Media

Groups

Tools

NTU CourseReserves ☒

Course Management

Control Panel

My Filing Cabinet

Course Tools

Evaluation

Grade Center

▼

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1. Copy + Paste + Steal: Artists Battle For Copyright vs Generative AI | Undercover Asia | Full Episode ▼



Copy + Paste + Steal: Artists Battle For Copyright vs Generative AI | Undercover Asia | Full Episode

Duration: 46:29

User: n/a - Added: 4/12/25

Watch Video

Artists & Generative AI Image Generation : Introduction & Current Developments.



2. Stable Diffusion explained (in less than 10 minutes) ▼



Stable Diffusion explained (in less than 10 minutes)

Duration: 9:56

User: n/a - Added: 3/29/24

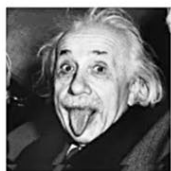
Watch Video

Diffusion-Based Generative AI Image Generation Technique - Stable Diffusion Explained (in 10 mins)

IT NEEDS A TRAINED MODEL



How can we create totally new images
?



Albert Einstein



2.3 billion images x 80KB = 184 Terabytes
How do they fit into a 2.13 GB model
?

Stable Diffusion 1.5: 2.3 billion image-text pairs
(from the LAION-5B dataset)

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Large-scale Artificial Intelligence Open Network

TRULY OPEN AI. 100% NON-PROFIT. 100% FREE.

LAION, as a non-profit organization, provides datasets, tools and models to liberate machine learning research. By doing so, we encourage open public education and a more environment-friendly use of resources by reusing existing datasets and models.

[Re-LAION 5B release \(30.08.2024\)](#)

LAION-400M

An open dataset containing 400 million English image-text pairs.

LAION-5B

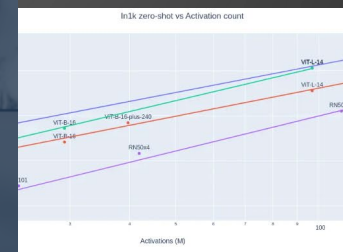
A dataset consisting of 5.85 billion multilingual CLIP-filtered image-text pairs.

Clip H/14

The largest CLIP (Contrastive Language-Image Pre-training) vision transformer model.

LAION-Aesthetics

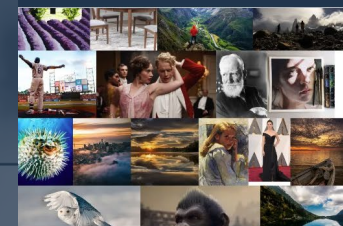
A subset of LAION-5B filtered by a model trained to score aesthetically pleasing images.



Large scale openCLIP: L/14, H/14 and g/14 trained on LAION-2B

by: Romain Beaumont, 15 Sep, 2022

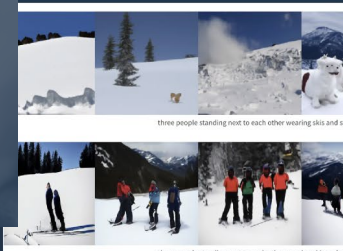
We trained three large CLIP models with OpenCLIP: ViT-L/14, ViT-H/14 and ViT-g/14 (ViT-g/14 was trained only for about a third the epochs compared to the rest). The H/14 model achieves 78.0% zero shot top-1 accuracy on ImageNet and 73.4% on zero-shot image retrieval at Recall@5 on MS COCO. As of Sep...



LAION-Aesthetics

by: Christoph Schuhmann, 16 Aug, 2022

We present LAION-Aesthetics, several collections of subsets from LAION 5B with high visual quality. To create LAION-Aesthetics we trained several lightweight models that predict the rating people gave when they were asked "How much do you like this image on a scale from 1 to 10?". LAION-Aesthetics ...



LAION-5B: A NEW ERA OF OPEN LARGE-SCALE MULTI-MODAL DATASETS

by: Romain Beaumont, 31 Mar, 2022

We present a dataset of 5,85 billion CLIP-filtered image-text pairs, 14x bigger than LAION-400M, previously the biggest openly accessible image-text dataset in the world - see also our NeurIPS2022 paper See our update on the LAION-5B dataset. Large image-text models like ALIGN, BASIC, Turing BletchL...

LAION-400-MILLION OPEN DATASET

by: Christoph Schuhmann, 20 Aug, 2021



word2vec

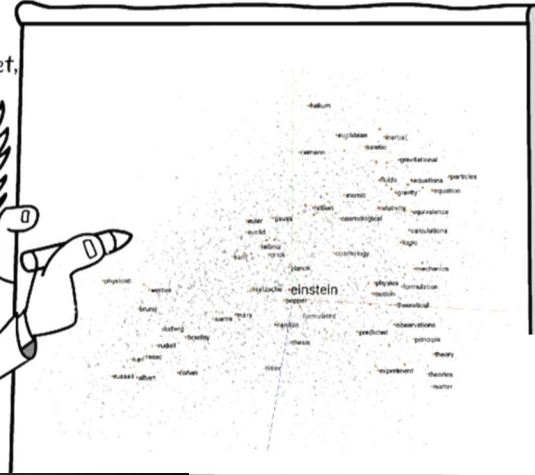
a	-0.32, 0.64, -0.49 ...
aardvark	0.62, -0.65, -0.47 ...
abaca	0.07, -1.00, 0.56 ...
abalone	-0.25, 0.66, 0.35 ...
abandon	0.61, 0.67, 0.99 ...

...

zucchini	0.62, -0.65, -0.47 ...
zucchini	0.07, -1.00, 0.56 ...
zugzwang	0.13, 0.98, -0.71 ...
zwanziger	-0.09, 0.44, -0.73 ...
twitter	-0.17, 0.43, 0.03 ...

All words of the English language are converted into vectors.
Words used in similar contexts, based on all written texts on the internet, are encoded into similar vectors.

Here's an example for Einstein



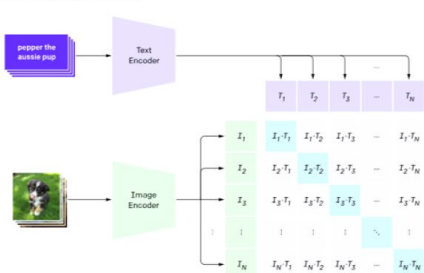
EMBEDDING THE IMAGE DESCRIPTIONS

"A painting of Mona Lisa"

the words

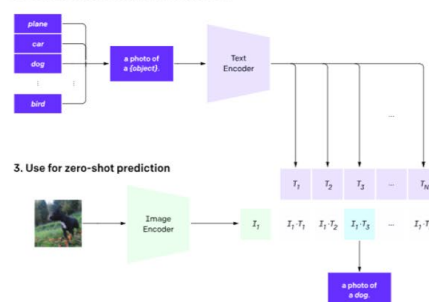
CLIP (Contrastive Language-Image Pre-Training)

1. Contrastive pre-training



Overview A

2. Create dataset classifier from label text

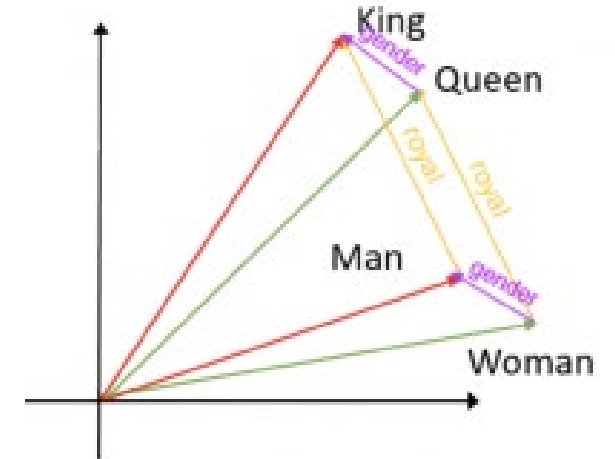


Overview B

CLIP pre-trains an image encoder and a text encoder to predict which images were paired with which texts in our dataset. We then use this behavior to turn CLIP into a zero-shot classifier. We convert all of a dataset's classes into captions such as "a photo of a dog" and predict the class of the caption CLIP estimates best pairs with a given image.

vectors, you can do calculations

man + woman = queen



IN LAYER

LATENT DIFFUSION MODELS

High-Resolution Image Synthesis with Latent Diffusion Models

Robin Rombach¹ Andreas Blattmann¹ Dominik Lorenz¹ Patrick Esser² Björn Ommer¹
¹Leibniz Universität of Munich & TUM, ²Heidelberg University, Germany ³Runway ML
<https://github.com/CompVis/latent-diffusion>

Abstract

By decomposing the image formation process into a sequential application of denoising autoencoders, diffusion models (DMs) achieve state-of-the-art synthesis results on image data and beyond. Additionally, their formulation allows for a guiding mechanism to control the image generation process without remaining. However, since these models typically operate directly in pixel space, optimization of powerful DMs often consumes hundreds of GPU days and inference is expensive due to sequential evaluations. To enable DM training on limited computational resources while retaining their quality and flexibility, we apply them in the latent space of powerful pretrained autoencoders. In contrast to previous work, training diffusion models on such a representation allows for the first time to reach a near-optimal point between complexity reduction and detail preservation, greatly boosting visual fidelity. By introducing cross-attention layers into the model architecture, we turn diffusion models into powerful and flexible generators for general conditioning inputs such as text or bounding boxes and high-resolution synthesis becomes possible in a convolutional manner. Our latent diffusion models (LDMs) achieve new state-of-the-art scores for image inpainting and class-conditional image synthesis and highly competitive performance on various tasks, including text-to-image synthesis, unconditional image generation and super-resolution, while significantly reducing computational requirements compared to pixel-based DMs.

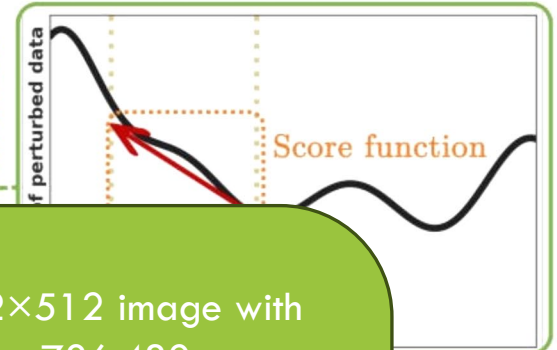
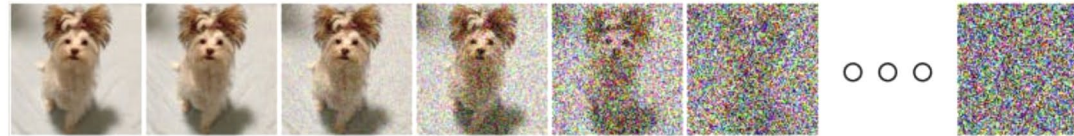
1. Introduction

Image synthesis is one of the computer vision fields with the most spectacular recent development, but also among those with the greatest computational demands. Especially high-resolution synthesis of complex, natural scenes is presently dominated by scaling up likelihood-based models, potentially containing billions of parameters in autoregressive (AR) transformers [6, 67]. In contrast, the promising results of GANs [1, 27, 40] have been revealed to be mostly confined to data with comparably limited variability as their adversarial learning procedure does not easily scale to modeling complex, multi-modal distributions. Recently, diffusion models [82], which are built from a hierarchy of denoising autoencoders, have shown to achieve impressive



Figure 1. Boosting the upper bound on achievable quality with less aggressive downsampling. Since diffusion models offer excellent inductive biases for spatial data, we do not need the heavy spatial downsampling of related generative models in latent space, but can still greatly reduce the dimensionality of the data via suitable autoencoding models, see Sec. 3. Images are from the DIV2K [1] validation set, evaluated at 512² px. We denote the spatial downsampling factor by f_s . Reconstruction FID₅₀ [29] and PSNR are calculated on ImageNet val. [13]; see also Tab. 8.

Data — Destructing data by adding noise — Noise



Data

The image space is enormous. Think about it: a 512×512 image with three color channels (red, green, and blue) is a 786,432-dimensional space! (You need to specify that many values for ONE image.)

Stable Diffusion is a latent diffusion model. Instead of operating in the high-dimensional image space, it first compresses the image into the **latent space ($4 \times 64 \times 64$)**. The latent space is 48 times smaller so it reaps the benefit of crunching a lot fewer numbers. That's why it's a lot faster.

LATENT DIFFUSION MODELS

High-Resolution Image Synthesis with Latent Diffusion Models

Robin Rombach¹ · Andreas Blattmann¹ · Dominik Lorenz¹ Patrick Esser² Björn Ommer¹

¹Ludwig-Maximilians-Universität München & TUM, Heidelberg University, Germany ²Runway ML
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By decomposing the image formation process into a sequential application of denoising autoencoders, diffusion models (DMs) achieve state-of-the-art synthesis results on image data and beyond. Additionally, their formulation allows for a guiding mechanism to control the image generation process without retraining. However, since these models typically operate directly in pixel space, optimization of powerful DMs often consumes hundreds of GPU days and inference is expensive due to sequential evaluations. To enable DM training on limited computational resources while retaining their quality and flexibility, we apply them in the latent space of powerful pretrained autoencoders. In contrast to previous work, training diffusion models on such a representation allows for the first time to reach a near-optimal point between complexity reduction and detail preservation, greatly boosting visual fidelity. By introducing cross-attention layers into the model architecture, we turn diffusion models into powerful and flexible generators for general conditioning inputs such as text or bounding boxes and high-resolution synthesis becomes possible in a convolutional manner. Our latent diffusion models (LDMs) achieve new state-of-the-art scores for image inpainting and class-conditional image synthesis and highly competitive performance on various tasks, including text-to-image synthesis, unconditional image generation and super-resolution, while significantly reducing computational requirements compared to pixel-based DMs.



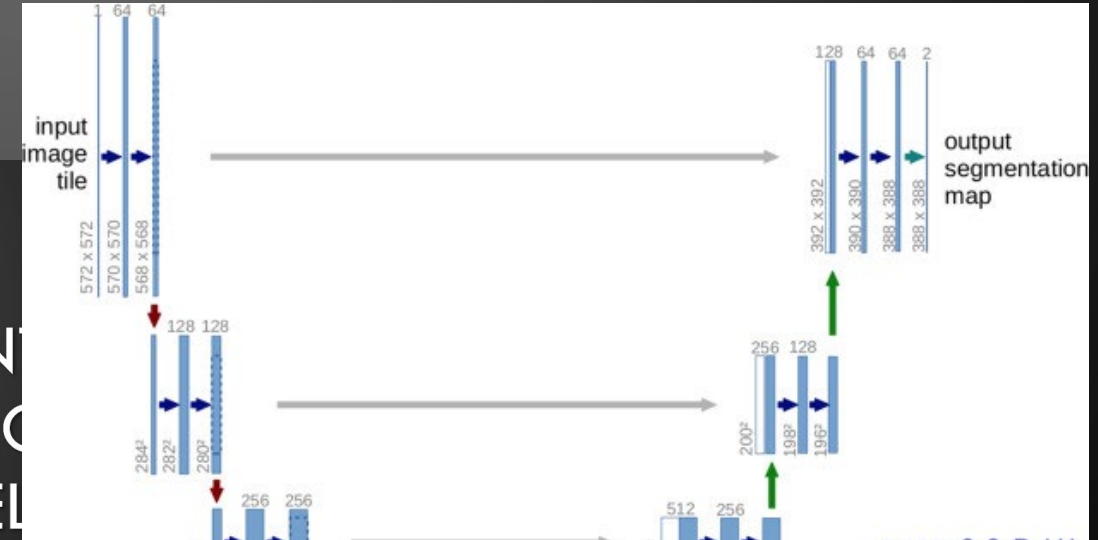
Figure 1. Boosting the upper bound on achievable quality with less aggressive downsampling. Since diffusion models offer excellent inductive biases for spatial data, we do not need the heavy spatial downsampling of related generative models in latent space, but can still greatly reduce the dimensionality of the data via suitable autoencoding models, see Sec. 3. Images are from the DIV2K [1] validation set, evaluated at 512² px. We denote the spatial downsampling factor by f . Reconstruction FIDs [29] and PSNR are calculated on ImageNet-val. [12]; see also Tab. 8.

results in image synthesis [36]. We define the state-of-the-art in class synthesis [15, 31] and super-resolution unconditional DMs can readily be seen as inpainting and colorization [85]. In contrast to other type models [19, 46, 69]. Being likelihood-based exhibit mode-collapse and training it and, by heavily exploiting parametric model highly complex distributions of out involving billions of parameters at.

Democratizing High-Resolution Image Synthesis We belong to the class of likelihood-based mode-covering behavior makes them excessive amounts of capacity (and thus on modeling imperceptible details of though the reweighted variational objective this by undersampling the initial are still computationally demanding evaluating such a model requires repetitions (and gradient computations) in the space of RGB images. As an example, training powerful DMs often takes hundreds of GPU-days (e.g. 150 · 1000 V100 days in [15]) and repeated evaluations on a noisy version of the input space render also inference expensive.

¹The first two authors contributed equally to this work.

LATENT DIFFUSION MODEL



Stable Diffusion is a text-to-image diffusion model, developed by CompVis at Ludwig-Maximilians-Universität München in the original 5 authors (Robin Rombach, Patrick Esser and Dominik Lorenz) and subsequent contributors.

the U-Net architecture is frequently used in diffusion models, particularly for image-based tasks. It's a convolutional neural network (CNN) with an encoder-decoder structure and skip connections, making it well-suited for tasks like image generation and denoising.

Each image gets converted into 3D tensors
(color value, width, height)
and fed into the U-Net step by step



Encoder

Each convolutional block extracts
more features of the image



With each encoding step the image gets downsampled,
so a larger part of the image with less details is extracted



Residual connections

Decoder

At decoding the image gets **upscaled** again,
to **consolidate** the informations gathered at encoding

The information gathered during each step of the encoding process
is combined with the results of decoding step
to **enhance the resolution while upscaling**

At the end of the encoding the U-Net has learned
WHAT is in the image, but not **WHERE**



That's been a tough one, I know!

INTRODUCING THE U-NET

CONDITIONAL DIFFUSION MODEL



Figure 8. Layout-to-image synthesis with an *LDM* on COCO [4], see Sec. 4.3.1. Quantitative evaluation in the supplement D.3.

4.3. Conditional Latent Diffusion

4.3.1 Transformer Encoders for LDMs

By introducing cross-attention based conditioning into LDMs we open them up for various conditioning modalities previously unexplored for diffusion models. For **text-to-image** image modeling, we train a 1.45B parameter *KL*-regularized *LDM* conditioned on language prompts on LAION-400M [78]. We employ the BERT-tokenizer [14] and implement τ_θ as a transformer [97] to infer a latent code which is mapped into the UNet via (multi-head) cross-attention (Sec. 3.3). This combination of domain specific experts for learning a language representation and visual synthesis results in a powerful model, which generalizes well to complex, user-defined text prompts, cf. Fig. 8 and 5. For quantitative analysis, we follow prior work and evaluate text-to-image generation on the MS-COCO [51] validation set, where our model improves upon powerful AR [17, 66] and GAN-based [109] methods, cf. Tab. 2. We note that applying classifier-free diffusion guidance [32] greatly boosts sample quality, such that the guided *LDM-KL-8-G* is on par with the recent state-of-the-art AR [26] and diffusion models [59] for text-to-image synthesis, while substantially reducing parameter count. To further analyze the flexibility of the cross-attention based conditioning mechanism we also train models to synthesize images based on **semantic layouts** on OpenImages [49], and finetune on COCO [4], see Fig. 8. See Sec. D.3 for the quantitative evaluation and implementation details.

Lastly, following prior work [3, 15, 21, 23], we evaluate our best-performing **class-conditional** ImageNet models with $f \in \{4, 8\}$ from Sec. 4.1 in Tab. 3, Fig. 4 and Sec. D.4. Here we outperform the state of the art diffusion model ADM [15] while significantly reducing computational requirements and parameter count, cf. Tab 18.

4.3.2 Convolutional Sampling Beyond 256^2

By concatenating spatially aligned conditioning information to the input of ϵ_θ , *LDMs* can serve as efficient general-

purpose image-to-image translation models. We use this to train models for semantic synthesis, super-resolution (Sec. 4.4) and inpainting (Sec. 4.5). For semantic synthesis, we use images of landscapes paired with semantic maps [23, 61] and concatenate downsampled versions of semantic maps with the latent image representation of our model (VQ-reg., see Tab. 8). We train on an input resolution of 256^2 (crops from 384^2) but find that our model generalizes to larger resolutions and can generate images up to megapixel regime when evaluated in a convolutional net (see Fig. 9). We exploit this behavior to also apply super-resolution models in Sec. 4.4 and the inpainting models in Sec. 4.5 to generate large images between 512×1024^2 . For this application, the signal-to-noise ratio (reduced by the scale of the latent space) significantly improves the results. In Sec. D.1 we illustrate this when learning *LDM* on (i) the latent space as provided by a $f = 8$ (KL-reg., see Tab. 8), and (ii) a rescaled version, scaled by the component-wise standard deviation.

The latter, in combination with classifier-free guidance [32], also enables the direct synthesis of > 2 megapixels for the text-conditional *LDM-KL-8-G* as in Fig.



Figure 9. A *LDM* trained on 256^2 resolution can generalize to larger resolution (here: 512×1024) for spatially conditioned tasks such as semantic synthesis of landscape images. See Sec. 4.3.2.

4.4. Super-Resolution with Latent Diffusion

LDMs can be efficiently trained for super-resolution by directly conditioning on low-resolution images via concatenation (cf. Sec. 3.3). In a first experiment, we follow SR3

A text-to-image diffusion model can be made conditional by encoding the text prompt into an embedding and feeding it to the denoising network. During the reverse diffusion process, the network uses this text embedding to guide the image generation, ensuring that the generated image matches the text description.

CONDITIONAL DIFFUSION MODEL



Figure 8. Layout-to-image synthesis with an LDM on COCO [4], see Sec. 4.3.1. Quantitative evaluation in the supplement D.3.

4.3. Conditional Latent Diffusion

4.3.1 Transformer Encoders for LDMs

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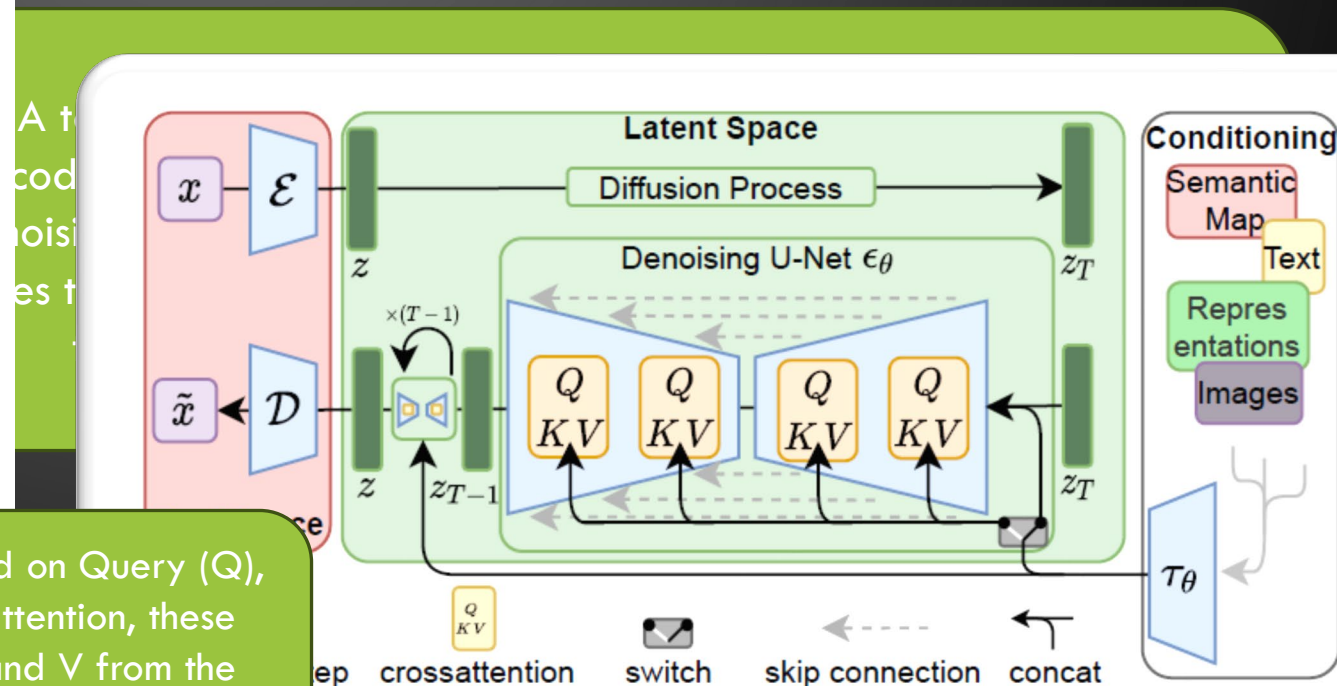
purpose image-to-image translation models. We use this to train models for semantic synthesis, super-resolution (Sec. 4.4) and inpainting (Sec. 4.5). For semantic synthesis, we use images of landscapes paired with semantic maps [23, 61] and concatenate downsampled versions of the semantic maps with the latent image representation of a $f = 4$ model (VQ-reg., see Tab. 8). We train on an input resolution of 256^2 (crops from 384^2) but find that our model generalizes to larger resolutions and can generate images up to the megapixel regime when evaluated in a convolutional manner (see Fig. 9). We exploit this behavior to also apply the super-resolution models in Sec. 4.4 and the inpainting models in Sec. 4.5 to generate large images between 512^2 and 1024^2 . For this application, the signal-to-noise ratio (induced by the scale of the latent space) significantly affects the results. In Sec. D.1 we illustrate this when learning an LDM on (i) the latent space as provided by a $f = 4$ model (KL-reg., see Tab. 8), and (ii) a rescaled version, scaled by the component-wise standard deviation.

The latter, in combination with classifier-free guidance [32], also enables the direct synthesis of $> 256^2$ images for the text-conditional LDM-KL-8-G as in Fig. 13.

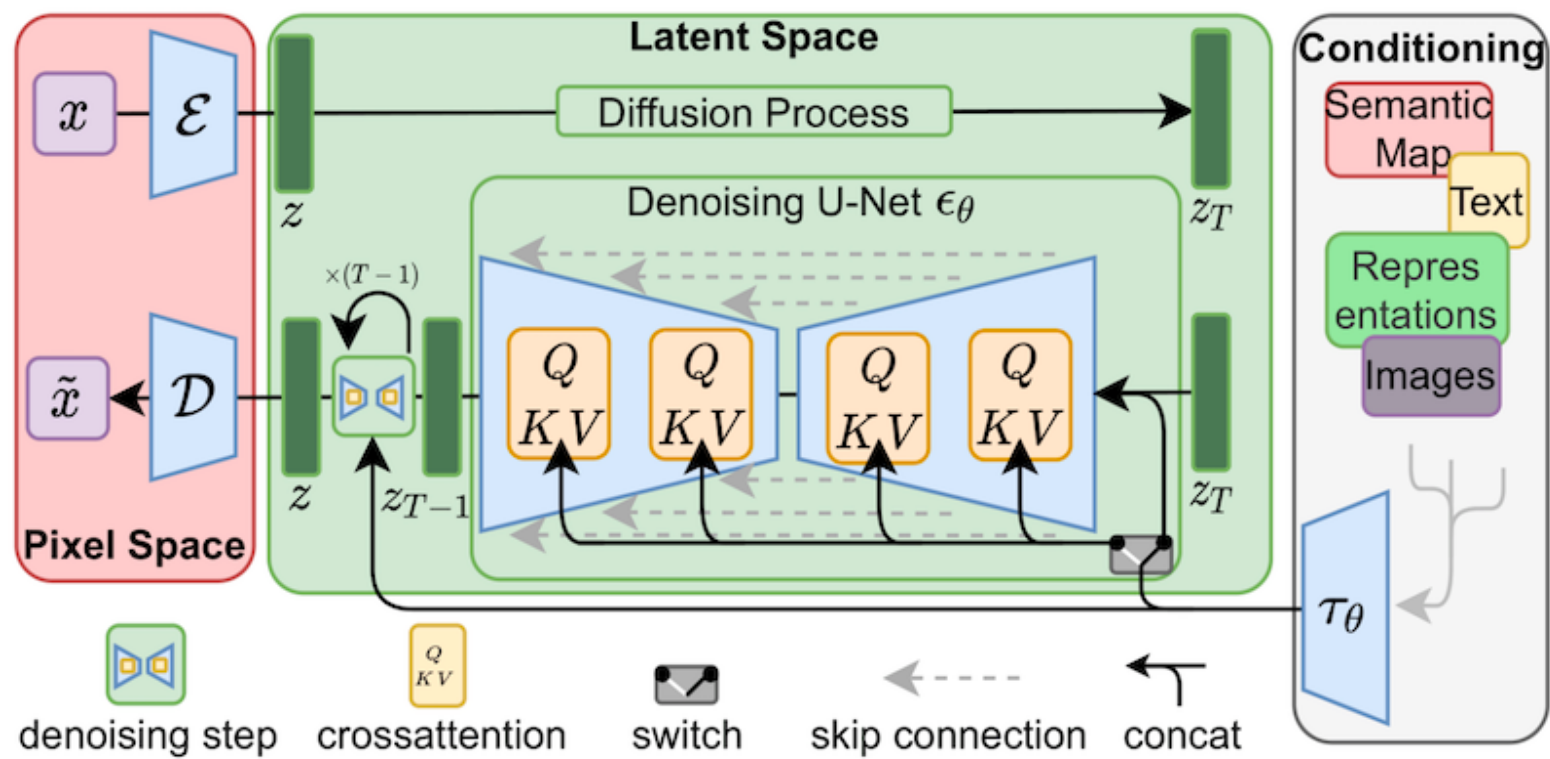
Cross-attention also computes attention scores based on Query (Q), Key (K), and Value (V) vectors. However, in cross-attention, these vectors are derived from different sequences. Q and V from the target sequence (decoder input), and K from the source sequence (encoder output)

By concatenating spatially aligned conditioning information to the input of ϵ_θ , LDMs can serve as efficient general-

directly conditioning on low-resolution images via concatenation (cf. Sec. 3.3). In a first experiment, we follow SR3



We condition LDMs either via concatenation or by a more general cross-attention mechanism. See Sec. 3.3



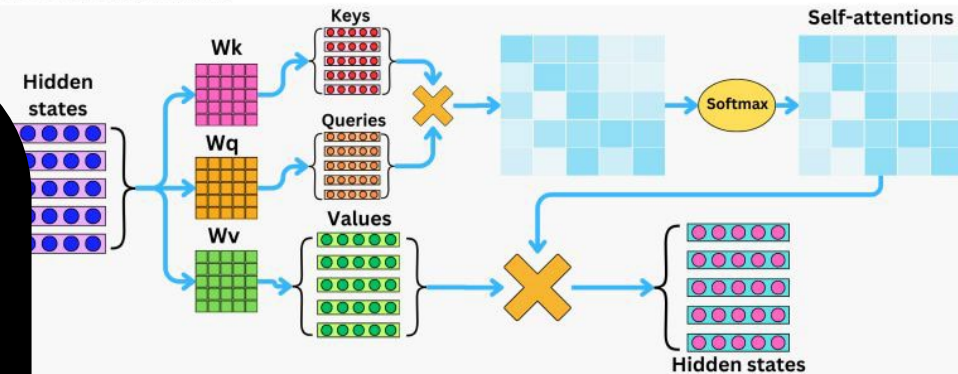
QKV

- The Query asks a question about a particular element.
- The Keys provide the labels of all other elements to be compared against the Query.
- The Values contain the actual information that is retrieved and combined based on the relevance scores derived from the Query-Key comparison, ultimately forming a richer, context-aware representation of the original element.

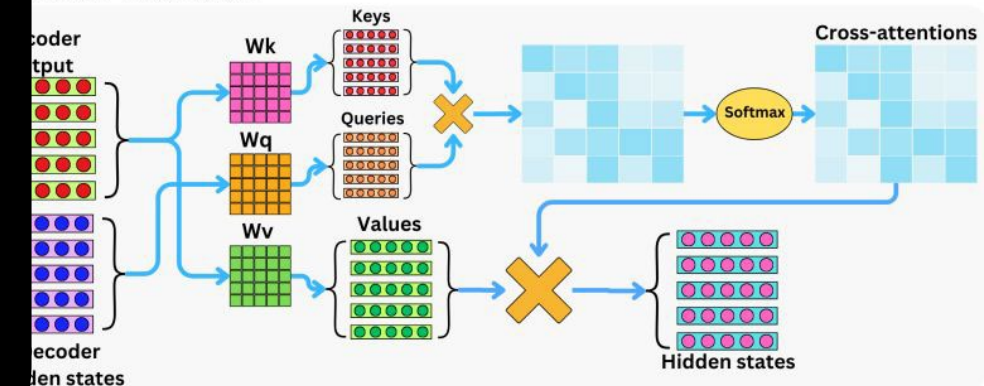
This QKV mechanism allows the model to selectively "pay attention" to different parts of the input sequence, enabling it to understand long-range dependencies and generate more meaningful representations of words or tokens within their specific contexts.

Self-Attention VS Cross-Attention

The Self-Attentions



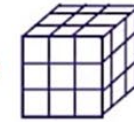
Cross-Attentions



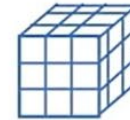
An image is encoded into an embedding vector by using the Convolutional layer



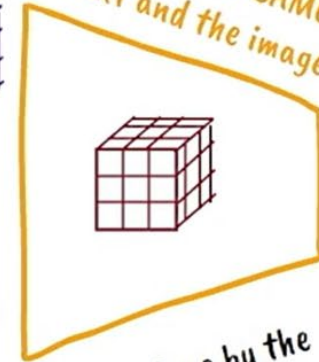
Convolutional layer



Self-Attention layer



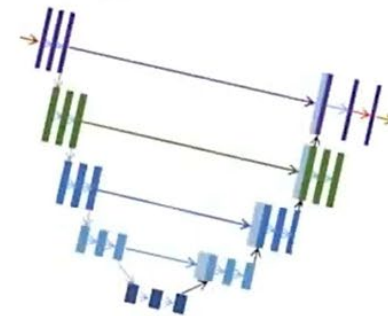
What if we could generate the **SAME** vector
for both, the text and the image?



It's done by the

CLIP-Encoder

and the encoded vector is
injected into the **U-Net**

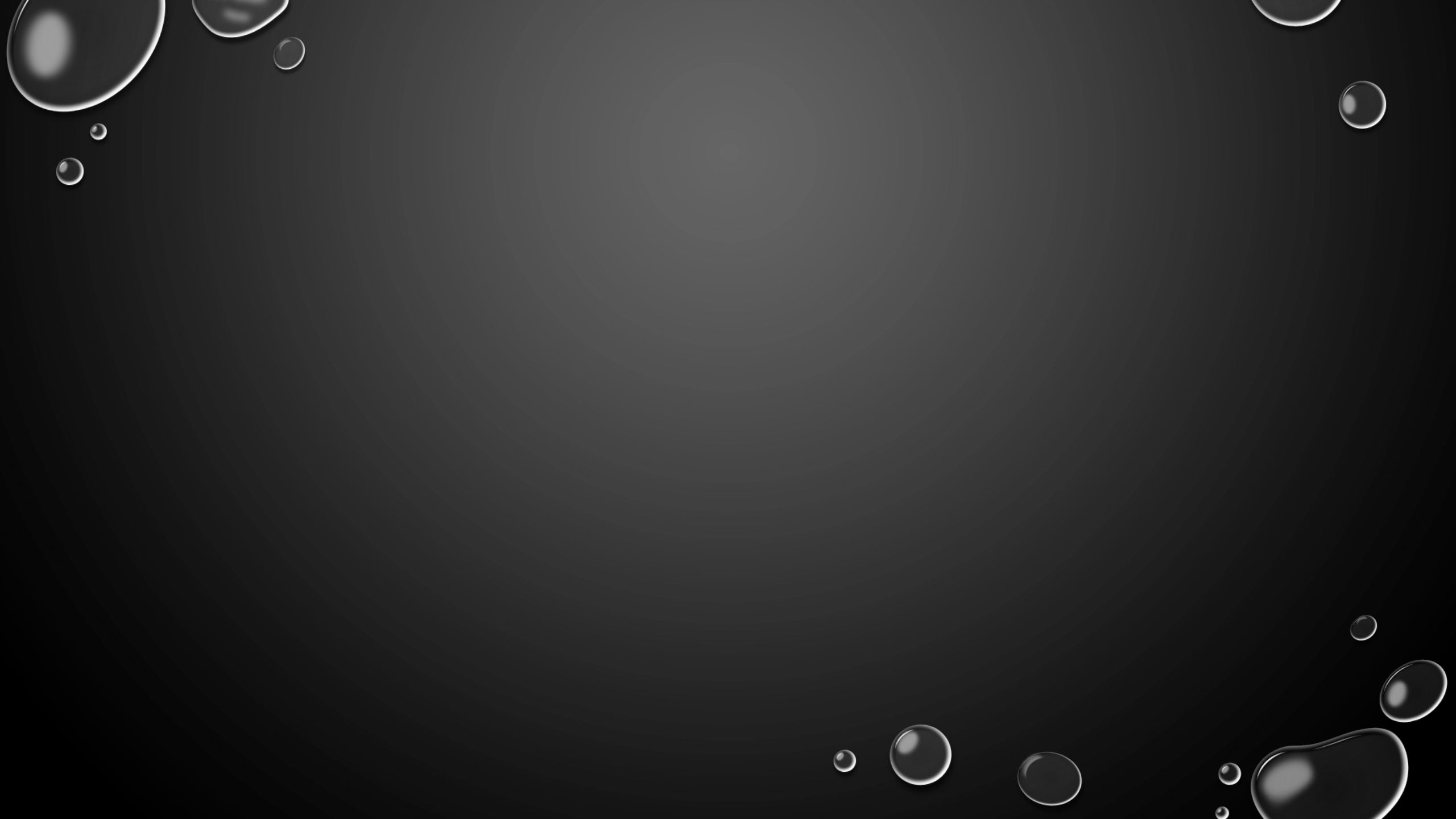


"A painting of Mona Lisa"

LET'S SUMMARIZE

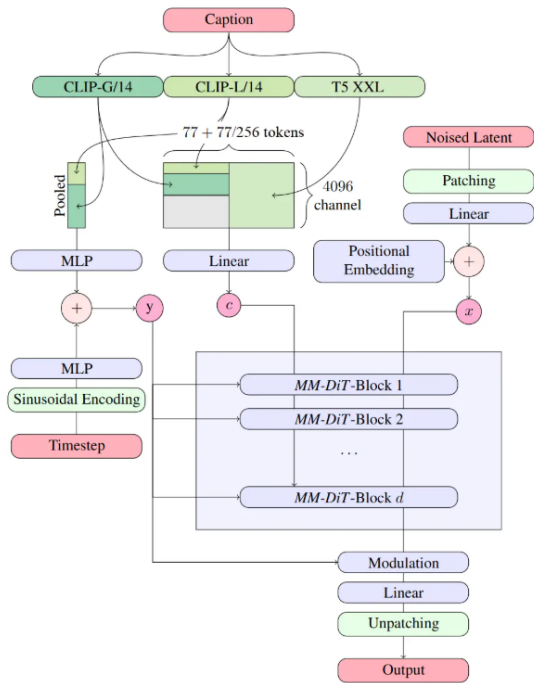


A text is encoded into an embedding vector by using the Self-Attention layer

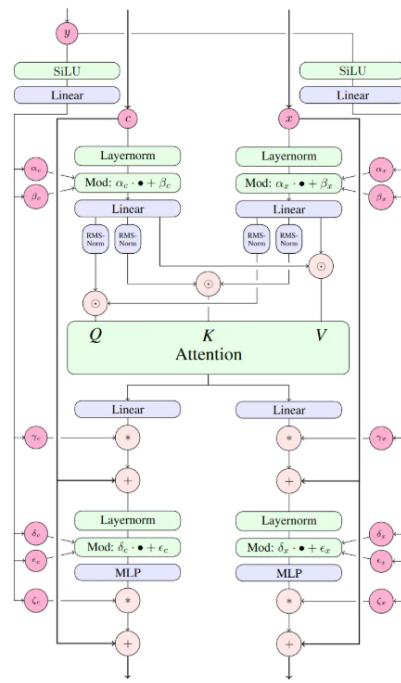


Stable Diffusion 3.5

It's not just random; it's a smart system. I learned it uses a mix of ideas called **Diffusion**, **Flow Matching**, and a special way to **sample time**. Let me try to explain it simply.



(a) Overview of all components.



(b) One MM-DiT block

How Stable Diffusion 3.5 Large Works

FLUX.1

FLUX.1 is designed to provide more control over the generated content, making it a powerful tool for artists, designers, and developers. FLUX.1 distinguishes itself by using Flow Matching and a DiT (Discrete Integrate and Transfer) architecture, similar to the advancements expected in models like Stable Diffusion 3.

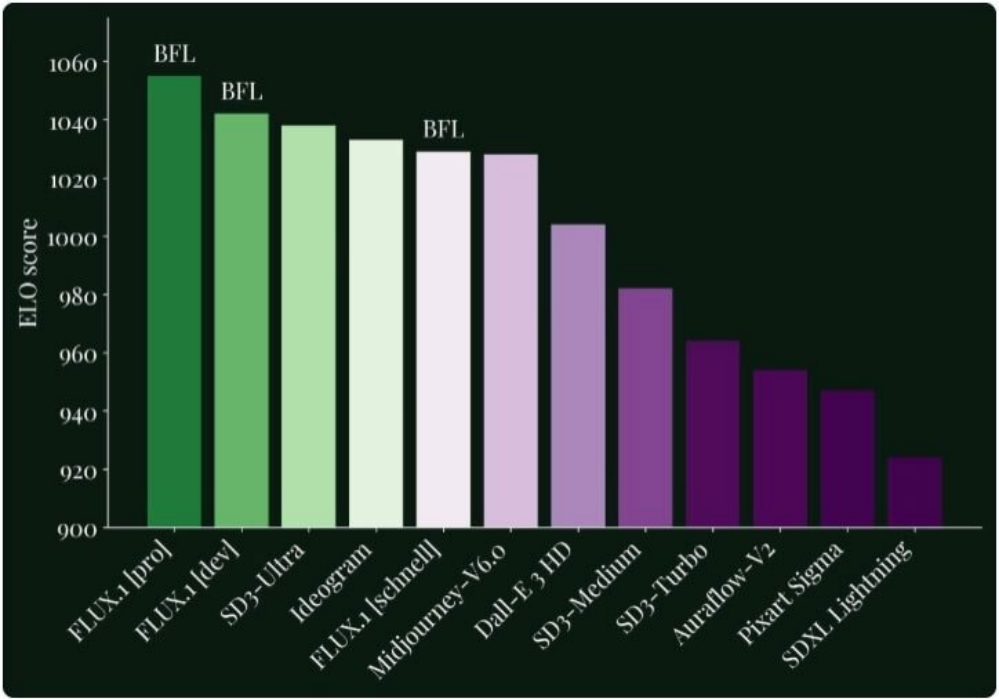
FLUX.1 is available in three variations:

- **FLUX Schnell:** This is a distilled version of FLUX.1 optimized for quick inference. It's ideal for applications where speed is crucial. However, it's not suitable for fine-tuning and is generally used for commercial purposes under an Apache license.
- **FLUX Dev:** This variant is tailored for fine-tuning. It provides the necessary flexibility and control for users who want to customize the model to their specific needs. A commercial license is required for this variant.
- **FLUX Pro:** The Pro version of FLUX.1 offers the highest quality and control. However, the weights are not open-source and are only accessible through an API. Fine-tuning is not supported on this variant.

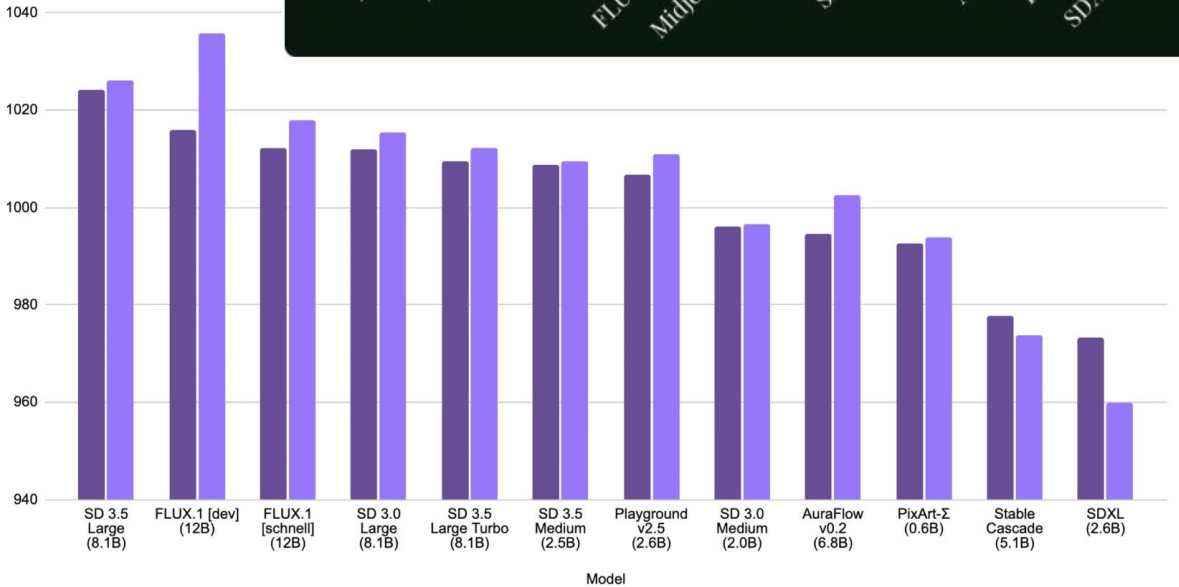
The specific version I used, Stable Diffusion 3.5 Large (with 8 billion parts, or parameters), uses an architecture called a **Multimodal Diffusion Transformer (MMDiT)**.

1. **Understanding My Words (Text Embeddings):** To understand my road scene prompts, it uses *three* different text "brains" (encoders like CLIP and T5). This helps it really get what I'm asking for. I can even turn off the biggest one (T5 XXL) to save computer memory if I need to.
2. **Chopping Up the Image (Patchify):** It takes the image (or its small latent version) and breaks it into little squares or "patches."
3. **The Main Engine (DiT Backbone):** The MMDiT uses a powerful Transformer (like those in ChatGPT) to process both the image patches and my text instructions together. It has special layers that pay attention to both what the image and what my words say.
4. **Efficient Math (adaLN):** It uses a smart kind of math (adaptive layer normalization) that works well and applies my text instructions even
5. **The Shrinker/Grower (VAE):** And, of course, it uses the VAE to down to the latent space and grow them back up.

This whole system lets Stable Diffusion 3.5 Large create detailed, 1-meg closely follow my complex text prompts.



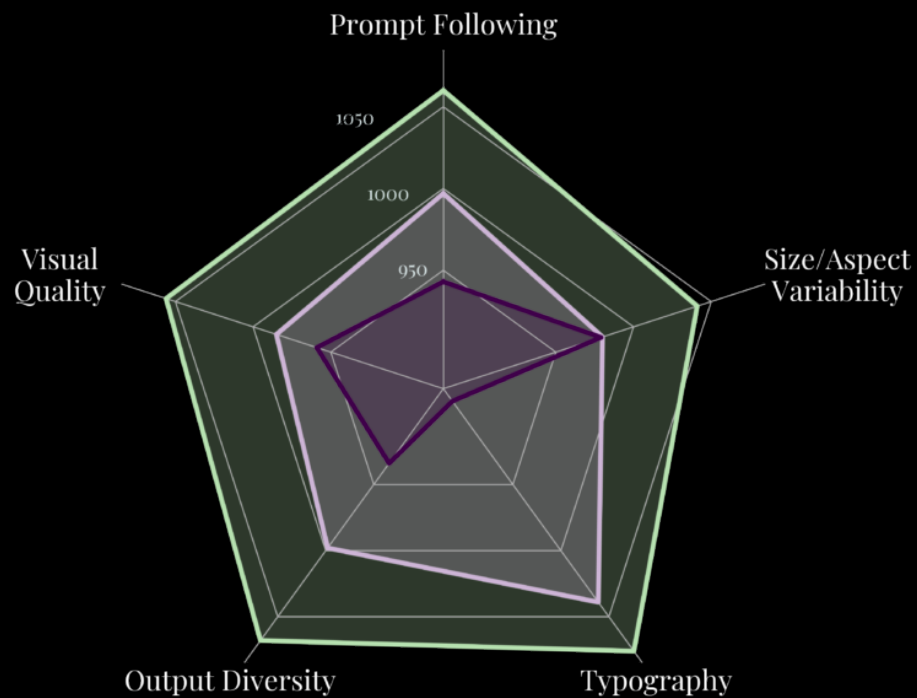
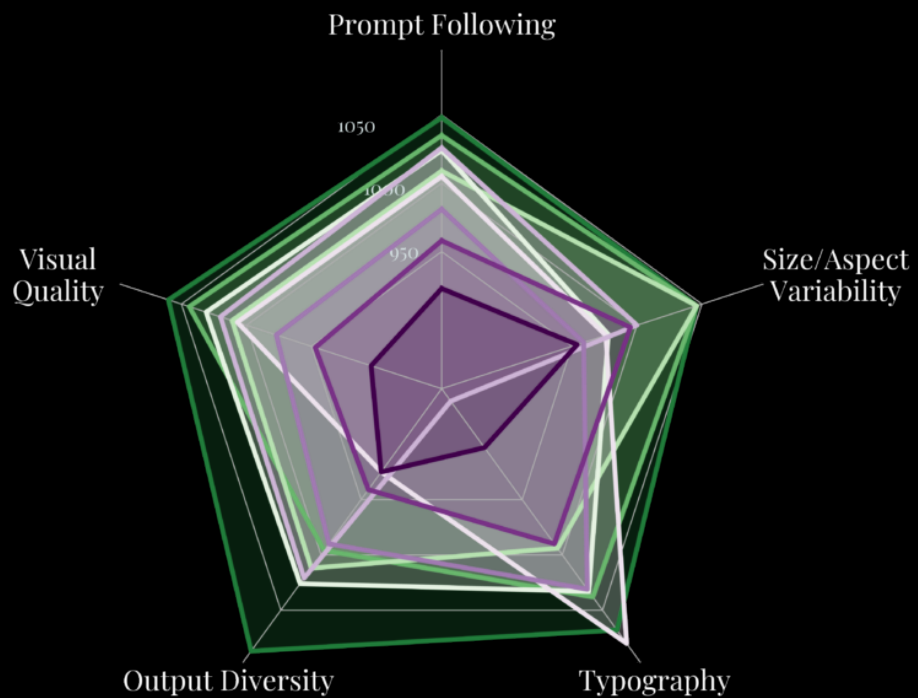
Prompt Adherence & Aesthetics



FLUX.1 [pro]
FLUX.1 [dev]
FLUX.1 [schnell]
SD3-Ultra
Ideogram

Midjourney-V6.o
Dall-E 3 HD
SD3-Medium
Auraflo-V2

FLUX.1 [schnell]
SD3-Turbo
SDXL-Lightning

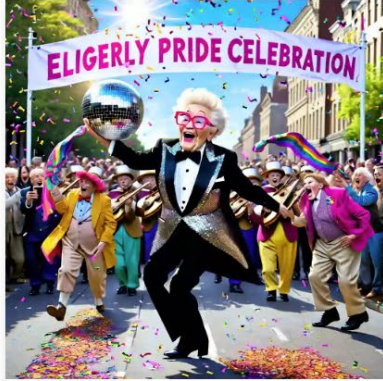


Prompt Description

a flamboyant octogenarian wearing a sequined tuxedo and oversized glasses enthusiastically dances down a packed city street, clutching an enormous disco ball in one hand while a marching band of elderly musicians follows closely behind, each member adorned in vibrant, mismatched outfits, the street is lined with astonished pedestrians capturing the eccentric parade on their smartphones, one onlooker is seen juggling brightly colored scarves, while a nearby mime dramatically mimics the dancer's moves, underneath a large banner reading "Elderly Pride Celebration," confetti rains down from overhead as the atmosphere buzzes with joy and laughter amidst the heartwarming chaos, the sun shines brightly against a clear blue sky, creating an uplifting and celebratory scene



Model: aurafLOW_v0.3



Model: aurafLOW



Model: fluxdev



Model: sd3l



Model: sd3m



Model: sdcascade



Model: sdxl



Model: dalle3

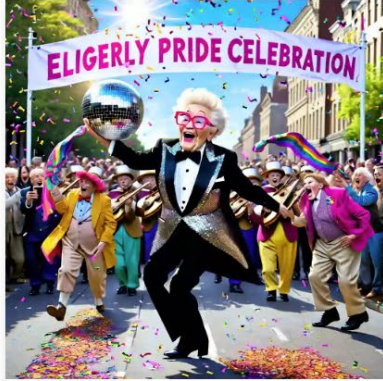


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Model: sd3l



Model: sd3m



Model: sdcascade



Model: sdxl



Model: dalle3





THE END

(3RD LAST SLIDE)